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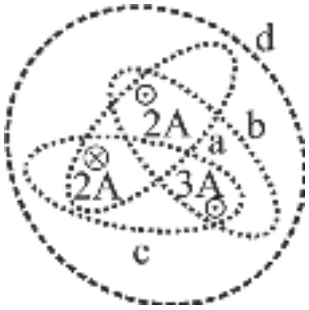
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# PHYSICS

1) Rank the value of $\oint \vec{B}.\vec{dl}$ for the closed paths shown in figure from the smallest to largest.



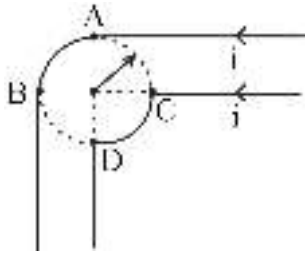

(1) a, b, c, d

(2) a, c, d, b

(3) a, d, c, b

(4) a, c, b, d

2) All straight wires are very long. Both AB and CD arc of the same circle, both subtending right angles at the centre O. Then the magnetic field at O is :-



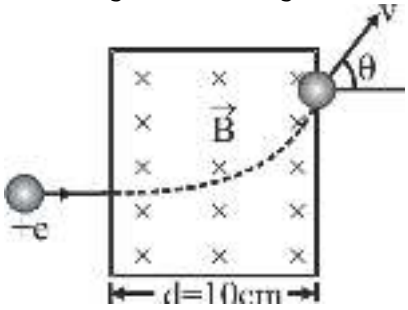

(1) $\frac{\mu_0 i}{4\pi R}$

(2) $\frac{\mu_0 i}{4\pi R}\sqrt{2}$

(3) $\frac{\mu_0 i}{2\pi R}$

(4) $\frac{\mu_0 i}{2\pi R}(\pi + 1)$

3) A proton accelerated by a potential difference 500 KeV moves through a transverse magnetic field of 0.51 T as shown in figure. The angle θ through which the proton deviates from the initial direction of its motion is



(1) 15°

(2) 30°

(3) 45°

(4) 60°

4)

An ionized gas contains both positive and negative ions. If it is subjected simultaneously to an electric field along the +x direction and a magnetic field along the +z direction, then :-

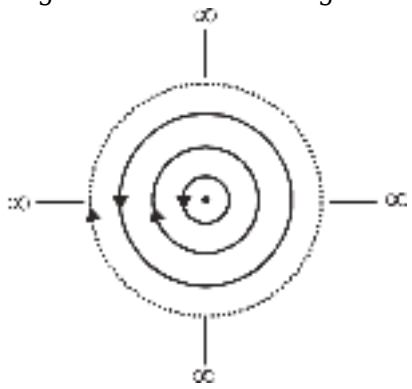
(1) Positive ions deflect towards +y direction and negative ions towards -y direction

(2) All ions deflect towards +y direction

(3) All ions deflect towards -y direction

(4) Positive ions deflect towards -y direction and negative ions towards +y direction

5) In given figure, infinite conducting rings each having current i in the direction shown are placed concentrically in the same plane as shown in the figure. The radii of rings are r, 2r, $2^2$r, $2^3$r, ....., .



The magnetic field at the centre of rings will be

(1) Zero

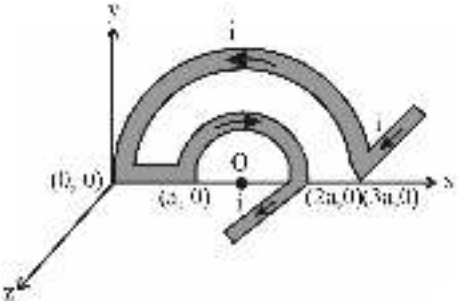
(2) $\frac{\mu_0 i}{r}$

(3) $\frac{\mu_0 i}{2r}$

(4) $\frac{\mu_0 i}{3r}$

6)

In the given figure net magnetic field at O will be :-



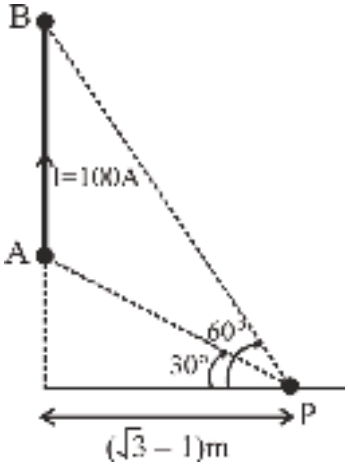

(1) $\frac{2\mu_0 i}{3\pi a}\sqrt{4-\pi^2}$

(2) $\frac{\mu_0 i}{3\pi a}\sqrt{4+\pi^2}$

(3) $\frac{2\mu_0 i}{3\pi a^2}\sqrt{4+\pi^2}$

(4) $\frac{2\mu_0 i}{3\pi a}\sqrt{(4-\pi^2)}$

7) A straight current carrying wire carries current 100 A, as shown in figure. The magnitude of magnetic field at point P which is at perpendicular distance $(\sqrt{3}-1)\,m$ from the wire if end A and end B of the element subtend angle 30° and 60° at point P, as shown is :



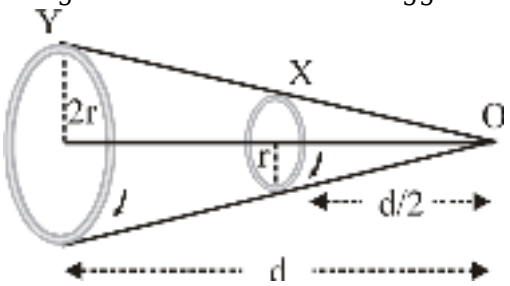

(1) $5 \times 10^{-6}$ T

(2) $2.5 \times 10^{-6}$ T

(3) $2.5 \times 10^{-5}$ T

(4) $8 \times 10^{5}$ T

8) Two circular coils X and Y, having equal number of turns, carry equal currents in the same sense and subtend same solid angle at point O. If the smaller coil X is midway between O and Y, and if we represent the magnetic induction due to bigger coil Y at O as $B_Y$ and that due to smaller coil X at O as $B_x$, then :-




(1) $\frac{B_Y}{B_X} = 1$

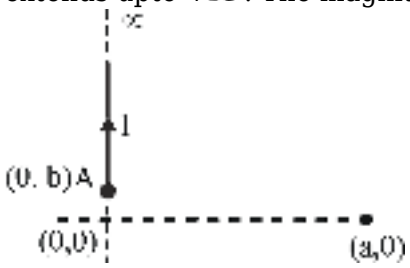
(2) $\frac{B_Y}{B_X} = 2$

(3) $\frac{B_Y}{B_X} = \frac{1}{2}$

(4) $\frac{B_Y}{B_X} = \frac{1}{4}$

9) A wire carrying current I is along y-axis such that its one end is at point A(0, b) while the wire

extends upto $+\infty$ . The magnitude of magnetic field strength at point (a, 0).



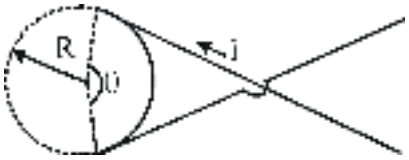

(1) $\frac{\mu_0 I}{4\pi a}\left(1+\frac{b}{\sqrt{a^2+b^2}}\right)$

(2) $\frac{\mu_0 I}{4\pi a}\left(1-\frac{b}{\sqrt{a^2+b^2}}\right)$

(3) $\frac{\mu_0 I}{4\pi a}\left(\frac{b}{\sqrt{a^2+b^2}}\right)$

(4) None of these

10) A wire carrying current i has the configuration shows in fig. Two semi-infinite straight sections, both tangent to the same circle, are connected by a circular arc, of central angle $\theta$, along the circumference of the circle, with all sections lying in the same plane. What must $\theta$ be in order for B to be zero at the centre of the circle?




(1) $\theta$ = 2 rad

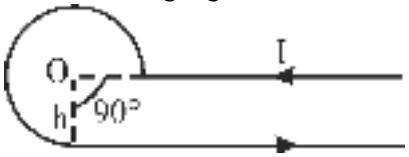
(2) $\theta$ = 10 rad

(3) $\theta$ = 5 rad

(4) $\theta$ = 2p rad

11)

What will be the value of the magnetic field induction at the centre of the coil, if I = 2A, h = 10 cm in the following figure?




(1) $1.14\times10^{-5}$T

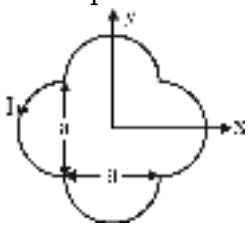
(2) $\frac{4\pi}{\mu_0}\times 1.14\times 10^{-5}$T

(3) $\frac{1.14}{4\pi}\times 10^{-5}$T

(4) $\frac{\mu_0}{4\pi}\times 2.14\times 10^{-5}$T

12)

A loop carrying current I lies in the x-y plane as shown in the figure. The unit vector $\hat{k}$ is coming out of the plane of the paper. The magnetic moment of the current loop is :




(1) $a^2 I\hat{k}$

(2) $\left(\frac{\pi}{2}+1\right) a^2 I\hat{k}$

(3) $-\left(\frac{\pi}{2}+1\right) a^2 I\hat{k}$

(4) $(2\pi+1)\, a^2 I\hat{k}$

13)

If the number of turns and current is doubled in a long solenoid keeping the length same, then magnitude of magnetic field at the centre of it becomes

(1) Four times

(2) Two times

(3) Half times

(4) Three times

14)

Which of the following statement is incorrect?

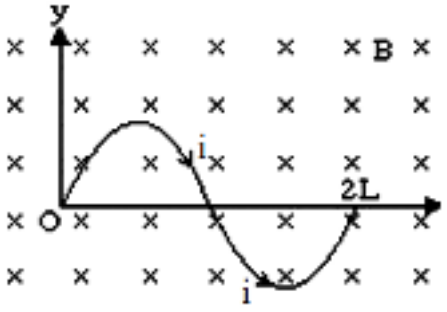
(1) A charge particle cannot be accelerated by uniform magnetic field

(2) When a charge particle projected in uniform magnetic field, then change in kinetic energy of the particle is zero

(3) Force acting on the charge particle in uniform magnetic field is always perpendicular to velocity

(4) Magnetic field is non-conservative force field

15) A wire carrying a current i is placed in a uniform magnetic field in the form of the curve




$y = a\sin\left(\frac{\pi x}{L}\right)$, $0 \leqslant x \leqslant 2L$. The force acting on the wire is

(1) $\frac{iBL}{\pi}$

(2) $iBL\pi$

(3) $2iBL$

(4) Zero

16) The magnetic field due to a small current element at a distanceand element carrying current i is,
or
Vector form of Biot-savart's law is

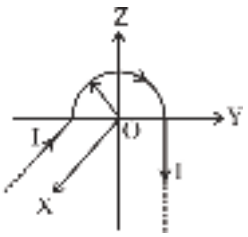
(1) $d\vec{B} = \frac{\mu_0}{4\pi} i \left( \frac{d\vec{l} \times \vec{r}}{r} \right)$

(2) $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left( \frac{d\vec{l} \times \vec{r}}{r} \right)$

(3) $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left( \frac{d\vec{l} \times \vec{r}}{r^2} \right)$

(4) $d\vec{B} = \frac{\mu_0}{4\pi} i \left( \frac{d\vec{l} \times \vec{r}}{r^3} \right)$

17) The magnetic field at the point 'O' is the radius of the curved part of the wire is R:-



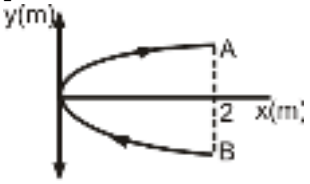

(1) $\frac{\mu_0 I}{4R}\left(1+\frac{1}{\pi}\right)\left(-\hat{i}\right)+\frac{\mu_0 I}{4\pi R}\left(-\hat{k}\right)$

(2) $\frac{\mu_0 I}{4R}\left(1+\frac{1}{\pi}\right)\left(\hat{i}\right)+\frac{\mu_0 I}{4\pi R}\left(-\hat{k}\right)$

(3) $\frac{\mu_0 I}{4R}\left(1-\frac{1}{\pi}\right)\left(-\hat{i}\right)+\frac{\mu_0 I}{4\pi R}\left(-\hat{k}\right)$

(4) None of these

18) A conducting wire bent in the form of a parabola $y^2 = 2x$ carries a current i = 2A as shown in figure. This wire is placed in a uniform magnetic field $\vec{B} = -4\hat{k}$ Tesla. The magnetic force on the wire is (in newton)



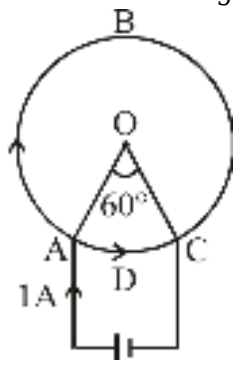

(1) $-16\hat{i}$

(2) $32\hat{i}$

(3) $-32\hat{i}$

(4) $16\hat{i}$

19) A cell is connected between the points A and C of a circular conductor ABCD with O as centre and angle AOC = 60°. If $B_1$ and $B_2$ are the magnitudes of the magnetic fields at O due to the currents in ABC and ADC respectively, then ratio $\frac{B_1}{B_2}$ is :-



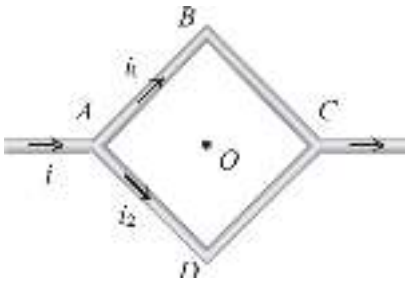

(1) 1

(2) 2

(3) 5

(4) 6

20) Figure shows a square loop ABCD with edge length a. The resistance of the wire ABC is R and that of ADC is 2R. The value of magnetic field at the centre of the loop assuming uniform wire is




(1) $\frac{\sqrt{2}\,\mu_0 i}{3\pi a}$ ⊙

(2) $\frac{\sqrt{2}\,\mu_0 i}{3\pi a}$ ⊗

(3) $\frac{\sqrt{2}\,\mu_0 i}{\pi a}$ ⊙

(4) $\frac{\sqrt{2}\,\mu_0 i}{\pi a}$ ⊗

21) A long hollow cylindrical wire carries a current I, having inner and outer radii 'R' and 2R respectively. Magnetic field at a point which 3R/2 distance away from its axis is :-

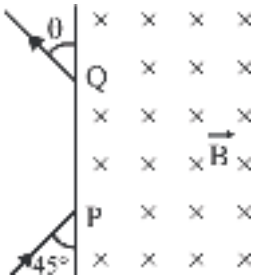
(1) $\frac{5\mu_0 I}{18\pi R}$

(2) $\frac{\mu_0 I}{36\pi R}$

(3) $\frac{5\mu_0 I}{36\pi R}$

(4) $\frac{5\mu_0 I}{9\pi R}$

22) A particle of mass $1.6 \times 10^{-27}$ kg and charge $1.6 \times 10^{-19}$ coulomb enters in a uniform magnetic field of 1 Tesla as shown in the fig. The speed of the particle is $10^7$ m/s. The distance PQ will be-




(θ = 45°)

(1) 0.14 m

(2) 0.28 m

(3) 0.4 m

(4) 0.5 m

23) The magnetic field existing in a region is given by $B = B_0\left[1 + \frac{x}{L}\right]\hat{k}$. A square loop of edge L and carrying current I is placed with its two edges parallel to x and y axes. The magnitude of net magnetic force experienced by loop is

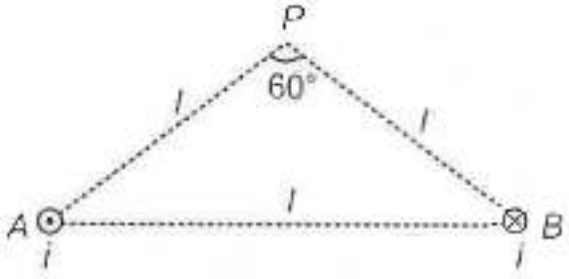
(1) $2B_0IL$

(2) Zero

(3) $B_0IL$

(4) $4B_0IL$

24) Figure shows two long wires A and B, each carrying a current i, separated by a distance l and oriented in a plane perpendicular to the plane of paper. The directions of currents are shown in figure. The magnitude of magnetic induction at a point P located at a distance l from both wires as shown in figure is




(1) $\frac{\mu_0 i}{8\pi l}$

(2) $\frac{\mu_0 i}{2\pi l}$

(3) $\frac{\mu_0 i}{2\pi l^2}$

(4) $\frac{\mu_0 i}{4\pi l}$

25) A charged particle of mass m and charge q is accelerated through a potential difference of V. It enters a region of uniform magnetic field B which is directed perpendicular to the direction of motion of the particle. The particle will move on a circular path of radius

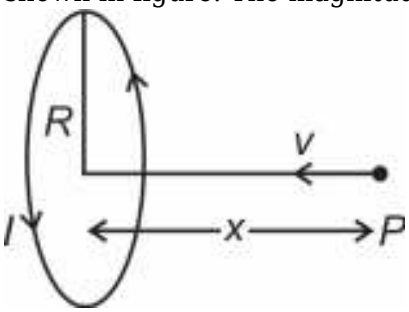
(1) $\sqrt{\frac{Vm}{2qB^2}}$

(2) $\frac{2Vm}{qB^2}$

(3) $\sqrt{\frac{2Vm}{q}}\left(\frac{1}{B}\right)$

(4) $\sqrt{\frac{Vm}{q}}\left(\frac{1}{B}\right)$

26) A proton moving with velocity v along the axis approaches a circular current carrying loop as shown in figure. The magnitude of magnetic force on proton at this instant is




(1) $\frac{\mu_0 evR^2}{2(R^2+x^2)^{3/2}}$

(2) $\frac{\mu_0 eviR^2}{2(R^2+x^2)^{3/2}}$

(3) $\frac{\mu_0 eviR^2}{4n(R^2+x^2)^{3/2}}$

(4) Zero

27) A proton moving with a constant velocity passes through a region of space without any change in its velocity. If E and B represent the electric and magnetic fields respectively, identify wrong statement

(1) E = 0, B = 0

(2) E = 0, B ≠ 0

(3) E ≠ 0, B = 0

(4) E ≠ 0, B ≠ 0

28) Radius of a current carrying coil is ‘R’. The ratio of magnetic field at a axial point which is R

distance away from the centre of the coil to the magnetic field at the centre of the coil :-

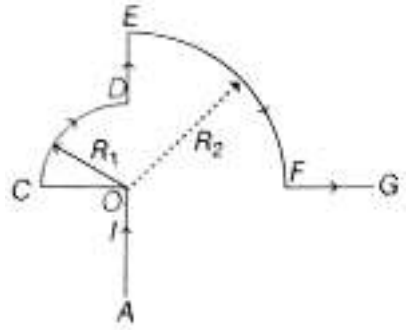
(1) $\left(\frac{1}{2}\right)^{1/2}$

(2) $\frac{1}{2}$

(3) $\left(\frac{1}{2}\right)^{3/2}$

(4) $\frac{1}{4}$

29) In the current carrying conductor (AOCDEFG) as shown, the magnetic induction at the point O is ($R_1$ and $R_2$ are radii of are CD and EF respectively. I = current in the loop, $\mu_0$ = permeability of free space)




(1) $\frac{\mu_0 I}{8}\left(\frac{R_1+R_2}{R_1R_2}\right)$

(2) $\frac{\mu_0 I}{8}\left(\frac{R_1-R_2}{R_1R_2}\right)$

(3) $\frac{\mu_0 I}{8}\left(\frac{R_1R_2}{R_1-R_2}\right)$

(4) $\frac{\mu_0 I}{8}\left(\frac{R_1R_2}{R_1+R_2}\right)$

30) An $\alpha$-particle and a proton, accelerated through same potential difference, enter into region of uniform magnetic field with their velocities perpendicular to the field. If radius of circular path traversed by proton is $\sqrt{2}$ cm, then radius of circular path traversed by $\alpha$-particle is.

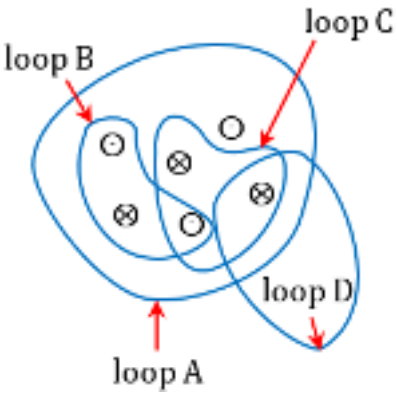
(1) 1 cm

(2) 2 cm

(3) 3 cm

(4) 4 cm

31) Consider six wires coming into or out of the page, all with the same current. Rank the line integral of the magnetic field (from most positive to most negative) considering the sense of




circulation in counterclockwise around each loop shown.

(1) B > C > D > A

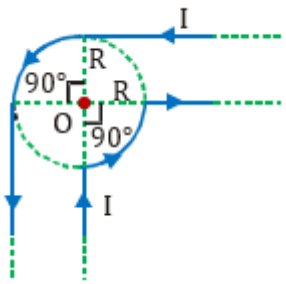
(2) B > C = D > A

(3) B > A > C = D

(4) C > B = D > A

32) Find out magnetic field at point 'O' for the following current distributions.

(b)




(1) $\frac{\mu_0 I}{8R}\odot$

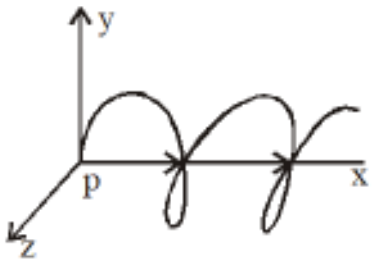
(2) $\frac{\mu_0 I}{4R}\left(1+\frac{2}{\pi}\right)\odot$

(3) $\frac{\mu_0 I}{4R}\odot$

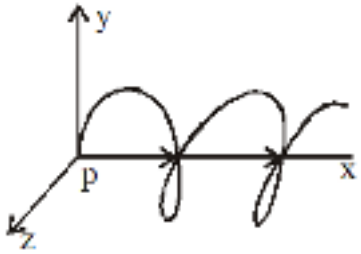
(4) $\frac{\mu_0 I}{4R}\otimes$

33) An electron is projected from origin with velocity $3.2\hat{i}+\hat{j}\,(m/s)$. A uniform magnetic field of $2\hat{k}$ tesla is present. The path followed by electron is correctly shown in ($m_e = 9 \times 10^{-31}$ kg)

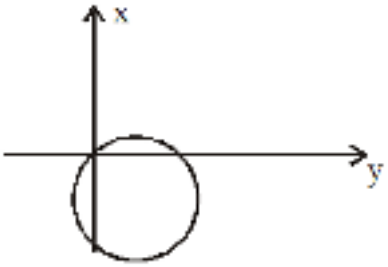
(1)



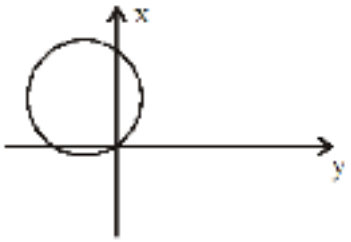

Helix on +ve side of z-axis

(2) 

Helix on -ve side of z-axis

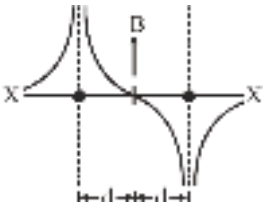
(3) 

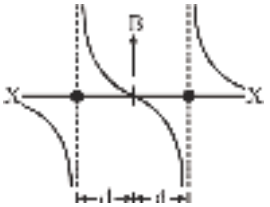
circle in xy plane

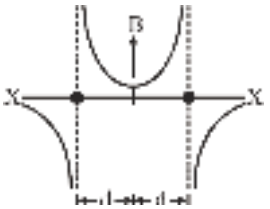
(4) 

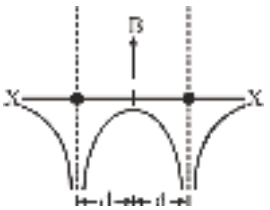
circle in xy plane

34) Two long parallel wires are at a distance 2d apart. They carry steady equal currents flowing out of the plane of the paper. The variation of the magnetic field B along the line XX' is given by :-

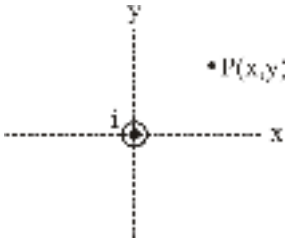
(1) 


(2) 


(3) 


(4) 

35) In a long wire current flowing along +z direction. The magnetic field at point P (x,y) is:-



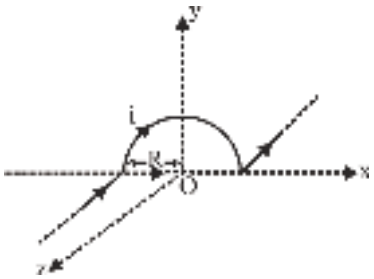

(1) $\frac{\mu_0 i}{2\pi\left(x^2+y^2\right)}\left(x\hat{i}-y\hat{j}\right)$

(2) $\frac{\mu_0 i}{2\pi\sqrt{x^2+y^2}}\left(x\hat{i}-y\hat{j}\right)$

(3) $\frac{\mu_0 i}{2\pi\sqrt{x^2+y^2}}\left(x\hat{j}-y\hat{i}\right)$

(4) $\frac{\mu_0 i}{2\pi\left(x^2+y^2\right)}\left(x\hat{j}-y\hat{i}\right)$

36) Magnetic field at point 'O' of given current distribution :-




(1) $\left(\frac{\mu_0 i}{4\pi R}\sqrt{\pi^2+4}\right)(\hat{k}+\hat{j})$

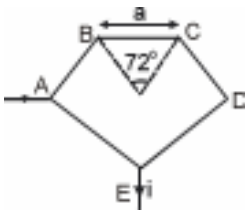
(2) $-\left(\frac{\mu_0 i}{4R}\right)\hat{k}$

(3) $\left(\frac{\mu_0 i}{4R}\right)\hat{k}$

(4) $\left(\frac{\mu_0 i}{4\pi R}\sqrt{\pi^2+4}\right)(\hat{k}-\hat{j})$

37)

Magnetic field strength at the centre of regular pentagon made of a conducting wire of uniform cross section area as shown in figure is :



(1) $\frac{5\mu_0 i}{4\pi a}\left[2\sin\frac{72^\circ}{2}\right]$

(2) $\frac{\mu_0 i}{\pi a}\left[2\sin\frac{72^\circ}{2}\right]$

(3) Not zero if current 'i' leaves D point instead of E

(4) Zero even if the current 'i' leaves point D instead of point E

38)

A coil having N turns is wound tightly in the form of a spiral with inner and outer radii 'a' and 'b' respectively. Find the magnetic field at centre, when a current I passes through coil:

(1) $\frac{\mu_0 I}{4(a-b)}\left[\frac{1}{a}-\frac{1}{b}\right]$

(2) $\frac{\mu_0 I}{8}\left[\frac{a+b}{a-b}\right]$

(3) $\frac{\mu_0 I N}{2(b-a)}\log_e\left(\frac{b}{a}\right)$

(4) $\frac{\mu_0 I}{8}\frac{a-b}{a+b}$

39) Three concentric thin spherical shells are of radii a, b and c (a < b < c). The first and third are connected by a fine wire through a small hole in the second. The second is connected to earth. Find the capacity of the condenser :

(1) $2\pi\varepsilon_0\left[\frac{a+b}{b-a}+\frac{c^2}{c-b}\right]$

(2) $4\pi\varepsilon_0\left[\frac{ab}{b-a}+\frac{c^2}{c-b}\right]$

(3) $4\pi\varepsilon_0\left[\frac{ab}{b+a}+\frac{c^2}{c+b}\right]$

(4) none of these

40) On placing a dielectric slab between the plates of a charged condenser its:-

| | **Capacity** | **Charge** | **Potential Difference** | **Energy stored** |
|---|---|---|---|---|
| (1) | decreases | remains unchanged | decreases | increases |
| (2) | increases | remains unchanged | increases | increases |

| | | | | |
|---|---|---|---|---|
| (3) | increases | remains unchanged | decreases | decreases |
| (4) | decreases | remains unchanged | increases | decreases |

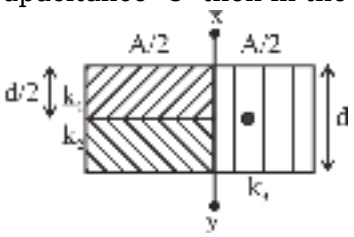
(1) 1

(2) 2

(3) 3

(4) 4

41) If air filled capacitor has capacitance 'C' then in the figure a capacitor is filled with dielectrics




the resultant capacitance is :-

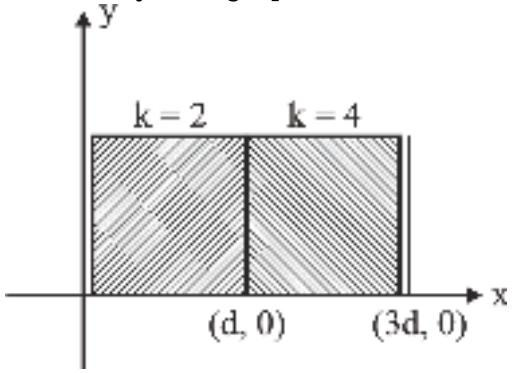
(1) $2C\left(\frac{1}{k_1}+\frac{1}{k_2}+\frac{1}{k_3}\right)$

(2) $C\left(\frac{1}{k_1}+\frac{1}{k_2}+\frac{1}{k_3}\right)$

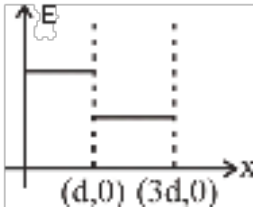
(3) $2C\,[k_1 + k_2 + k_3]$

(4) $C\left(\frac{k_1k_2}{k_1+k_2}+\frac{k_3}{2}\right)$

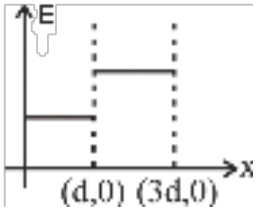
42) A parallel plate capacitor has two layers of dielectric as shown in figure. This capacitor is connected across a battery. The graph which shows the variation of electric field (E) and distance (x) from left plate :



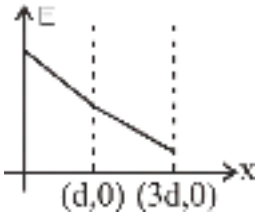

(1)



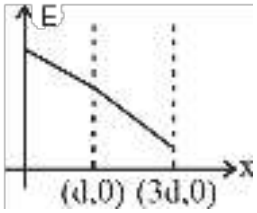

(2)



(3)




(4)




43)

The capacity of a parallel plate condenser is $C_0$. If a dielectric of relative permitivity $\varepsilon_r$ and thickness equal to one fourth the plate separation is placed between the plates, then its capacity becomes C.

The value of $\frac{C}{C_0}$ will be:-

(1) $\frac{5\in_r}{4\in_r+1}$

(2) $\frac{4\in_r}{3\in_r+1}$

(3) $\frac{3\in_r}{2\in_r+1}$

(4) $\frac{2\in_r}{\in_r+1}$

44) Separation between the plates of a parallel plate capacitor is d and the area of each plate is A. When a slab of material of dielectric constant k and thickness t(t<d) is introduced between the plates, its capacitance becomes :-

(1) $\frac{\varepsilon_0 A}{d+t\left(1-\frac{1}{k}\right)}$

(2) $\frac{\varepsilon_0 A}{d+t\left(1+\frac{1}{k}\right)}$

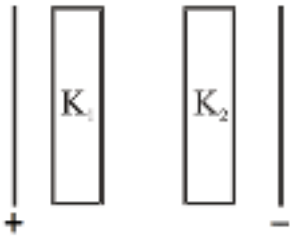
(3) $\frac{\varepsilon_0 A}{d-t\left(1-\frac{1}{k}\right)}$

(4) $\frac{\varepsilon_0 A}{d-t\left(1+\frac{1}{k}\right)}$

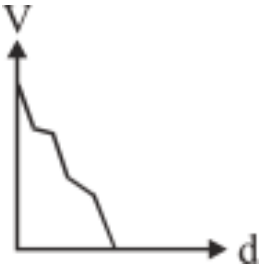
45)

A charged parallel plate capacitor is given. Choose most appropriate graph of potential of it. ($K_1 > K_2$)

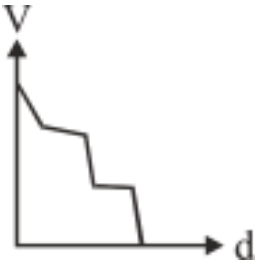
(Where K represent dielectric constant)



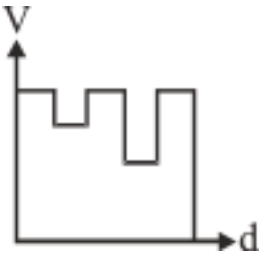

(1)



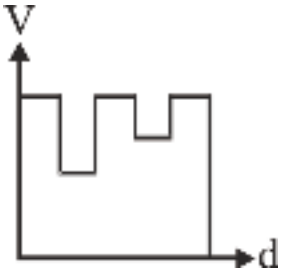

(2)




(3)




(4)




## CHEMISTRY

1) $K_6\left[(CN)_5Co-O-O-Co(CN)_5\right] \xrightarrow{\text{Oxidises}}$

(x)

$K_5\left[(CN)_5Co-O-O-Co(CN)_5\right]$

(y)

In both complexes Co have $t_2g^6 eg^0$ configuration the bond energy (B.E.) in x and y is-

(1) B.E. of (O–O) in y < B.E. of (O–O) in x

(2) B.E. of (O–O) in x < B.E. of (O–O) in y

(3) B.E. of (O–O) in x = B.E. of (O–O) in y

(4) B.E. of (O–O) in x and B.E. of (O–O) in y can't be predicted.

2) A, B and C are possible complexes of $CoCl_3.6H_2O$. A shows maximum electrical conductance and C

has maximum percentage of nonionisable $Cl^-$. Then incorrect statements is:

(1) No weight loss, when A is treated with conc.$H_2SO_4$

(2) Order of number of moles of AgCl ppt on adding excess $AgNO_3$ solution is A > B > C

(3) Complex C is $[Co(H_2O)_3Cl_3].3H_2O$

(4) Complex B is $[Co(H_2O)_5Cl]Cl_2.H_2O$

3) Which of the following shows maximum number of total isomers ?

(1) $[Co(en)_3]^{3+}$

(2) $[Pt(NH_3)(H_2O)(Cl)(Br)]$

(3) $[Pd(NH_3)_4(NO_2)_2]$

(4) $[Zn(NH_3)_2(Cl)_2]$

4) Match the column :-

| | | | |
|---|---|---|---|
| (A) | $Na_4[Fe(CN)_5NOS]$ | (P) | PV = +2 |
| (B) | cis-Platin | (Q) | SV = 4 |
| (C) | EDTA | (R) | Flexidentate ligand |
| (D) | Prussian blue | (S) | Hexacyano complex of iron |

| | A | B | C | D |
|---|---|---|---|---|
| (1) | S | R | Q | P,R |
| (2) | P | P,Q | R | S |
| (3) | Q,P | R | S | P |
| (4) | P | R | R,S | Q |

(1) (1)

(2) (2)

(3) (3)

(4) (4)

5) Choose incorrect stability order :

(1) $[Cu(NH_3)_4]^{2+} < [Cu(en)_2]^{2+} < [Cu(trien)]^{2+}$

(2) $[Fe(H_2O)_6]^{3+} < [Fe(NO_2)_6]^{3-} < [Fe(NH_3)_6]^{3+}$

(3) $[Co(H_2O)_6]^{3+} < [Rh(H_2O)_6]^{3+} < [Ir(H_2O)_6]^{3+}$

(4) $[Cr(NH_3)_6]^{1+} < [Cr(NH_3)_6]^{2+} < [Cr(NH_3)_6]^{3+}$

6) A blue colouration is obtained when :-

(1) $NH_4OH$ is added to $CuSO_4$

(2) $FeCl_3$ reacts with $K_4[Fe(CN)_6]$

(3) Anhydrous white $CuSO_4$ is dissolved in water

(4) All of these

7) The overall complex dissociation equilibrium constant for the complex $[Cu(NH_3)_4]^{2+}$ ion will be ($\beta_4$ for this complex is $2.1 \times 10^{13}$)

(1) $4.7 \times 10^{-14}$

(2) $2.1 \times 10^{13}$

(3) $11.9 \times 10^{-2}$

(4) $2.1 \times 10^{-13}$

8) Low spin tetrahedral complexes are not formed because

(1) for tetraheral complexes, the splitting energy is lower than pairing energy.

(2) for tetraheral complexes, the splitting energy is higher than pairing energy.

(3) electrons do not go to $e_g$ in case of tetrahedral complexes

(4) tetrahedral complexes are formed by weak field ligands only.

9)

Which of the following compounds show optical Isomerism :-

(i) Cis-$[Co(NH_3)_4Cl_2]^+$ (ii) trans-$[Co(en)_2Cl_2]^+$
(iii) Cis-$[Co(en)_2Cl_2]^+$ (iv) $[Co(en)_3]^{3+}$

(1) (i) & (ii)

(2) (ii) & (iii)

(3) (iii) & (iv)

(4) (i), (ii) & (iv)

10)

| | **Compound** | | **Total stereoisomers** |
|---|---|---|---|
| (A) | $[Co(en)_2Br_2]^+$ | (M) | 2 |
| (B) | $K_3[Co(OX)_3]$ | (N) | 3 |
| (C) | $[Pt(NH_3)_2Cl_2]$ | (O) | 4 |
| (D) | $[Pt(NH_3)_3Br_3]^+$ | (P) | 5 |

Correct is :

| | (A) | (B) | (C) | (D) |
|---|---|---|---|---|
| (1) | O | M | O | O |
| (2) | N | M | M | M |
| (3) | M | O | M | M |
| (4) | N | N | M | M |

(1) 1

(2) 2

(3) 3

(4) 4

11) In which of the following molecule orbital of different principle quantum number are used in the hybridization :-

(1) $IF_7$

(2) $[Ni(CO)_4]$

(3) $[PtCl_4]^{-2}$

(4) $K_4[Fe(F_6)]$

12) Select the correct M–C, Bond Strength?

(1) $[Ni(CO)_4] > [Co(CO)_4]^- > [Fe(CO)_4]^{-2}$

(2) $[Ni(CO)_4] < [Co(CO)_4]^- < [Fe(CO)_4]^{-2}$

(3) $[Ni(CO)_4] < [Co(CO)_4]^- = [Fe(CO)_4]^{-2}$

(4) $[Ni(CO)_4] = [Co(CO)_4]^- = [Fe(CO)_4]^{-2}$

13) Which of the following d-orbital used in the complex $[Ni(CN)_5]^{-3}$ having atleast 4 identical metal ligand bond lengths ?

(1) $dx^2 - y^2$

(2) $dz^2$

(3) dxy

(4) dyz

14) Which of the following compound is Hetroleptic complex compound

(1) Zeise's salt

(2) Ferrocene

(3) Cromocene

(4) all of these are hetroleptic complex compounds

15) Which one of the following complexes will have six isomers?(Including steroisomers)

(1) $[Co\,(en)\,(NH_3)_2Cl_2]\,Cl$

(2) $[Cr(H_2O)_4Cl_2]\,Cl$

(3) $[Co(ox)_3]^{3-}$

(4) $[Co(en)_2Br_2]\,Cl$

16) Which of the following statement is correct?

(1) With $d^2sp^3$ hybridization $[FeCl(CN)_4(O_2)]^{4-}$ complex is diamagnetic

(2) $[NiCl_4]^{2-}$ complex is more stable than $[Ni(dmg)_2]$

(3) $[V(CO)_6]$ is not very stable and easily reduces to $[V(CO)_6]^-$

(4) Ligands such as CO, $CN^-$, $NO^+$ are pie– donar due to the presence of filled π-molecular orbital

17) A brown ring is formed in the ring test of $NO_3^-$ ion. It is due to the formation of

(1) $[Fe(H_2O)_5(NO)]^{2+}$

(2) $FeSO_4.NO_2$

(3) $[Fe(H_2O)_4(NO)_2]^{2+}$

(4) $FeSO_4.HNO_3$

18) **Statement I :** Toxic metal ions are removed by the chelating ligands.
**Statement II :** Chelate complexes tend to be more stable.

(1) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.

(2) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.

(3) Statement I is true; Statement II is false.

(4) Statement I is false; Statement II is true.

19) The coordination number of $Ni^{+2}$ is = 4
$NiCl_2 + KCN\,(Complex) \rightarrow A\,(Complex)$
A + Conc.HCl (Maximum) $\rightarrow$ B(Chloro complex)
The IUPAC name of A and B are

(1) Potassium tetracyanonickelate(II), potassium tetrachloronickelate(II)

(2) Tetracyanopotassiumnickelate(II), tetrachloro potassiumnickelate (II).

(3) Tetracyanonickel (II), tetrachloronickel (II)

(4) Ptassiumtetracyanonickel (II), potassiumtetrachloronickel (II).

20) In $Fe(CO)_5$, the Fe - C bond possesses

(1) Ionic character

(2) $\sigma$ - character only

(3) $\pi$ - character only

(4) Both $\sigma$ and $\pi$ character

21) The strongest CO bond is present in

(1) $[Cr(CO)_6]^+$

(2) $[Fe(CO)_5]$

(3) $[V(CO)_6]^-$

(4) All have equal strength

22) Correct relation between pairing energy (P) and splitting energy ($\Delta_0$) in complex ion $[Pt(H_2O)_6]^{+4}$

:–

(1) $\Delta_0 < P$

(2) $\Delta_0 > P$

(3) $\Delta_0 = P$

(4) Cannot be predicted

23) For the complex, $[Fe^{II}(en)_3]_n\ [Co^{III}(NH_3)_2(NO_2)_4]_2$ the value of n is :-

(1) 1

(2) 2

(3) 3

(4) 4

24)

Which of the following order is not correct :-

(1) $[Fe(H_2O)_6]^{+3} < [Fe(CN)_6]^{-3}$ $\Delta_0$ order

(2) $[Fe(CN)_6]^{-4} < [Fe(CN)_6]^{-3}$ Fe-C Bond length

(3) $CN^- <$ CO π-accepting tendency

(4) $C_2H_4 > K[PtCl_3(C_2H_4)]$ C-C bond length

25) "Crystal field splitting energy" (CFSE = "$\Delta_0$") for $[Ti(H_2O)_6]^{3+}$ is 242 kJ/mol. Then the "Crystal field stabilisation energy" in kJ/mol will be

(1) 96.80 kJ/mol

(2) –96.80 kJ/mol

(3) 242 kJ/mol

(4) –242 kJ/mol

26) $\left[Fe(en)_2(H_2O)_2\right]^{+2} + en \rightarrow$ complex (X). the correct statement about the complex (X) is

(1) it is a high spin complex

(2) it is diamagnetic

(3) it shows geometrical isomerism

(4) it is outer orbital complex

27) Which is correct :-

| **Complex** | **No. of Chilate rings** |
|---|---|
| (1) $[Co(ox)_2Cl_2]^{-3}$ | 4 |
| (2) $[Ni(dmg)_2]$ | 4 |
| (3) $[Ca(EDTA)]^{-2}$ | 6 |
| (4) Brown ring complex | 1 |

(1) 1

(2) 2

(3) 3

(4) 4

28) Which of the following is most likely formula of platinum complex, if $\frac{1}{4}$ of total chlorine of the compound is precipitated by adding $AgNO_3$ to its aqueous solution

(1) $PtCl_4.6H_2O$

(2) $PtCl_4.5H_2O$

(3) $PtCl_2.2H_2O$

(4) $PtCl_4.3H_2O$

29) If CO ligands are substituted by NO in respective neutral carbonyl compounds then which of the following will not be correct formula?

(1) $Cr(CO)_3(NO)_2$

(2) $Fe(CO)_2(NO)_2$

(3) $Cr(NO)_4$

(4) $Ni(CO)_2(NO)_2$

30) When 0.1 mol $CoCl_3(NH_3)_5$ is treated with excess of $AgNO_3$, 0.2 mol of AgCl are obtained. The conductivity of solution will correspond to :

(1) 1 : 3 electrolyte

(2) 1 : 2 electrolyte

(3) 1 : 1 electrolyte

(4) 3 : 1 electrolyte

31) Hybrid state of Aluminium in acidified aqueons solution of $AlCl_3$ is :-

(1) $sp^3$

(2) $sp^3d$

(3) $sp^3d^2$

(4) $sp^3d^3$

32) What is the following statements is not correct regarding $NO_2$ molecule :-

(1) Paramagnetic behaviour decreases when it undergoes in dimerisation

(2) It is coloured in its dimeric form

(3) The colour is due to the presence of unpaired electron

(4) The free electron is present in one of the $sp^2$ hybrid orbital

33) Which can displace Al from Alum :-

(1) Mn

(2) Tl

(3) Sc

(4) In

34) Which is not monomer for a high molecular mass silicone polymer.

(1) $PhSiCl_3$

(2) $MeSiCl_3$

(3) $Me_2SiCl_2$

(4) $Me_3SiCl$

35) Strong reducing behaviour of $H_3PO_2$ is due to :-

(1) High oxidation state of phosphorus

(2) Presence of two –OH groups and one P–H bond

(3) Presence of one –OH group and two P–H bonds

(4) High electron gain enthalpy of phosphorus

36) Which produce alkaline solution when dissolved in water
(i) Borax (ii) Potash alum
(iii) $SO_2$ (iv) $SO_2(OH)_2$

(1) (i) and (iii)

(2) (i) and (ii)

(3) (i), (ii) and (iii)

(4) Only (i)

37) $H_2C_2O_4 \xrightarrow{\Delta}$ gas(A) + gas(B) + liquid(C). Gas(A) burns with a blue flame and is oxidised to gas(B) Gas(A) + $Cl_2 \rightarrow$ (D). A, B and D are :

(1) $CO_2$, CO, $H_2O$

(2) CO, $CO_2$, $CO_2Cl_2$

(3) CO, $H_2O$, $NH_2CONH_2$

(4) CO, $CO_2$, $COCl_2$

38) How many S–S bonds, S–O–S bonds, σ bonds, π-bonds are present in trimer of sulphur trioxide:-

(1) 0, 3, 16, 2

(2) 0, 3, 12, 6

(3) 0, 6, 12, 16

(4) 0, 4, 12, 6

39) Which of the following statements is not correct ?

(1) $(CH_3)_3$ COH is less acidic than $(CH_3)_3$ SiOH

(2) Like CO, its analogue of Si is not stable

(3) Bond length [B – F] $BF_3 < BF_4^-$

(4) None of these

40) Consider the following statements.
$S_1$ : Fluorine does not form any polyhalide because it does not have d-orbitals in valence shell.
$S_2$ : In $ClF_3$, the three lone pairs of electrons occupy the equatorial position.
$S_3$ : $B_2Cl_6$ does not exist.
Of these statements.

(1) $S_1$, $S_2$ and $S_3$ are correct

(2) $S_1$ and $S_2$ are correct

(3) $S_1$ and $S_3$ are correct

(4) $S_2$ and $S_3$ are correct

41) Inorganic benzene ($B_3N_3H_6$) -

(1) Only 6 ($sp^2$ – $sp^2$) σ bonds and 3 pπ–pπ coordinate bonds

(2) 12 ($sp^2$ – $sp^2$) s bonds and 3 pπ–pπ coordinate bonds

(3) 6 ($sp^2$ – $sp^2$) σ bonds , 6 ($sp^2$ – σ) σ bonds

(4) 6 ($sp^2$ – $sp^2$) σ bonds , 6 ($sp^2$ – s) σ bonds and 3 pπ–pπ coordinate bonds

42) The common product of hydrolysis of $XeF_2$ & $XeF_6$ is/are :-

(1) Xe, $O_2$

(2) $XeO_3$, $O_2$

(3) Xe, $XeO_3$

(4) Only HF

43)

Find the ratio of $sp^2$ and $sp^3$ hybridised boron atom in $Na_2B_4O_7.10H_2O$

(1) 2 : 19

(2) 2 : 2

(3) 2 : 11

(4) 1 : 4

44) P–O–P bond is absent in :-

(1) Dimetaphosphoric acid

(2) Pyro phosphorous acid

(3) Cyclic trimetaphosphoric acid

(4) Hypo phosphorous acid

45) Correct statement for Graphite is :-

(1) C-C bond distance is nearly equal to distance between adjacent layers

(2) It can be used as a lubricant but not at very high temperature

(3) It has a layered struture with $sp^2$ hybridised Carbon

(4) It has more density than diamond

## BIOLOGY

1) (i) Thymus (ii) Spleen
(iii) Appendix (iv) MALT
(iv) Bone marrow
Which of the above are not secondary lymphoid organs ?

(1) iii, iv only

(2) i and v

(3) i, ii, iii. iv. v

(4) i, ii, iii, iv only

2) Which of the following sets of disease is caused by bacteria?

(1) Typhoid fever and mumps

(2) Pneumonia and Rabies

(3) Plague and small pox

(4) Tetanus and cholera

3) Antitoxin is a :-

(1) Preparation containing antibodies to the toxin

(2) Antigen which has the ability to generate immune response against a toxin.

(3) Preparation which provide passive immunisation.

(4) Both 1 & 3

4) Gametogony in life cycle of Plasmodium parasite :

(1) Starts in human RBCs and complete in mosquito stomach

(2) Occur in human liver and RBCs

(3) Starts in human RBCs and complete in mosquito RBCs

(4) Occurs in mosquito only

5) Which is incorrect statement :

(1) IgM – Protect body blood stream

(2) IgD – Activation of β - lymphocyte

(3) Fever includes in $1^{st}$ line of defence

(4) B - lymphocyte includes in $3^{rd}$ line of defence

6) A patient complaining of ache in the abdomen visited the doctor. The doctor following observed signs
I. Internal bleeding
II. Fever
III. Muscular pain
IV. Anemia
V. Blockage of intestinal passage
The patient is most likely suffering from

(1) Filariasis

(2) Amoebic dysentery

(3) Ringworm

(4) Ascariasis

7) Which of the following statements are not correct regarding immune system of higher vertebrates ?
(A) The human immune system consists of lymphoid organs, tissues, cells and insoluble fat molecules like antibodies.
(B) Memory based acquired immunity is characteristics of higher vertebrates.
(C) The antibodies are found in blood and they generate humoral immune response.
(D) The primary lymphoid organs are organs which provide the sites for interaction of lymphocytes with the antigen.
(E) Lymph nodes serve to trap the micro-organism or antigen which are present in tissue fluid.

(1) D and E

(2) Only D

(3) A and D

(4) A, D and E

8) Read the following four statements A, B, C and D which are the description of events in the life cycle of *Plasmodium vivax*. Select the right option having both the correct statemetns.
(A) Parasites reproduce asexually in the red blood cells, bursting the red blood cells and causing cycles of fever and other symptoms.
(B) Sexual stages gametocytes develop in stomach of mosquito.
(C) Fertilisation and development take place in the GUT of mosquito.
(D) Female mosquito takes up gametes with blood meal.

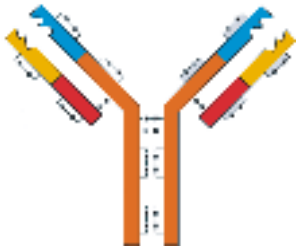
(1) A and B

(2) B and C

(3) A and C

(4) C and D

9) How many of below are incorrect about given diagram ?



(a) Gives AMI
(b) T-cells themselves do not secrete but help B-cell produce them
(c) $H_2L_2$ molecule
(d) Glycolipid molecule
(e) Antigen binding site called as epitope

(1) one
(2) two
(3) three
(4) four

10) Choose the option which describes the physical barrier of innate immunity.

(1) Acid in the stomach, saliva in the mouth, tears from eyes prevent microbial growth.
(2) Mucus coating of the epithelium lining the respiratory, gastrointestinal and urinogenital tracts help in trapping microbes entering our body.
(3) Chemical like interferons protects cells from viral infection.
(4) Rejection of an allograft.

11) Mark the correct statement :-
(a) Yoga has been practised to achieve physical and mental health
(b) Infectious diseases are very common and everyone of us suffers from these at sometime or other
(c) AIDS is not congenital
(d) Cancers are non-infectious
(e) Healthy persons bring economic prosperity

(1) a, b and c
(2) b, c, d and e
(3) c and d
(4) a, b, c, d and e

12) How many of the following statements are correct ?
A. Both bone-marrow and thymus provide micro-environments for the maturation of T-lymphocytes.
B. Bone-marrow and Thymus are lymphoid tissue.
C. Antigens trapped in lymph nodes are responsible for activation of lymphocytes present there and cause immune response.
D. T-suppressor cells suppress the functions of T-killer and T-helper cells.
E. IgG act as B-cell receptor and is involved in activation of B-lymphocytes.

(1) A, B and C
(2) A, C and D

(3) C and D

(4) A and C

13) How many of the following are bacterial diseases?
Pneumonia, Common cold, Plague, Malaria, Tetanus, AIDS, Typhoid, Diptheria

(1) Seven

(2) Six

(3) Five

(4) Four

14) Note the following words.

| | |
|---|---|
| I. Skin | II. Phagocytes |
| III. B-cells | IV. Inflammation |
| V. Antibodies | VI. T-cells |
| VII. Fever | VIII. Antimicrobial drugs |
| IX. NK-cells | X. Secretions |

Identify the factors involved in second line of defence :-

(1) II, IV, VII and IX

(2) II, III, V and IX

(3) IV, VI, VIII and X

(4) III, V, VII and VIII

15) A person is suffering from one disease and shows the following symptoms, like sustained high fever (39 to 40 degree celsius), weakness, stomach pain, constipation, headache and loss of appetite. His doctor confirmed that the he is suffering from one bacterial infection.
On which of the following facts (given below) his doctor confirmed the bacterial infection.:-
(i) Salmonella typhi bacteria are responsible for the infection.
(ii) Streptococcus pneumonia is responsible for the infection.
(iii) By seeing the report which shows the positive result of Widal test.
(iv) The pathogen is transmitted through the bite of female Anopheles mosquito.
(v) As a result of infection, the alveoli get filled with fluid leading to severe problem in respiration.

(1) (i) and (iii) only

(2) (ii), (iii) and (v) only

(3) (ii), (iii), (iv) and (v) only

(4) (i), (ii), (iv) and (v) only

16) Read the following statements (A-D).
(1) Memory-based innate immunity evolved in higher vertebrates based on the ability to differentiate foreign organism (e.g. pathogens) from self - cells.
(2) The exaggerated response of the immune system to certain allergen present in the environment is called allergy.
(3) Passive immunity is slow and takes time to give its full effective response.
(4) The Yellowish fluid colostrum secreted by mother during the initial days of lactation has abundant antibodies to protect the infant.
How many of the above statements are false ?

(1) 4

(2) 3

(3) 2

(4) 1

17) Abdominal spasm, vomiting, internal bleeding and intestinal blockage are caused by which of the following :-

(1) *Wuchereria bancrofti*

(2) Common round worm

(3) *Plasmodium falciparum*

(4) Toga virus

18)

Active artificial immunization is induced by administration of :-
(a) Bacterial products (b) Toxoids

(c) Dead microbes (d) Antitoxins

(1) a & b only

(2) a, b & c only

(3) b & d only

(4) c & d only

19) Study carefully the following stages of life cycle of malarial parasite i.e., Plasmodium. Arrange these stages in the correct sequence and select the correct answer.
1. Sporozoites leave the blood stream and enter the liver cells of man.
2. Sporozoites present in the salivary glands of female Anopheles mosquito are injected into the blood stream of man.
3. The parasite reproduces asexually in RBCs, resulting in bursting of RBCs and causing the cycles of fever, released parasites infect new RBCs.
4. The parasite reproduces asexually in liver cells, ultimately causing the rupturing of cells.
5. Gametocytes develop in the RBCs of man.
6. Female Anopheles mosquito takes up the gametocytes with blood meal of an infected person.
7. Mature infective stage of the parasite i.e., sporozoites escape from intestine and migrate to the mosquito's salivary glands.
8. Fertilization and developmental stages of the parasite take place in mosquito's stomach.

(1) 2 → 1 → 4 → 3 → 5 → 6 → 8 → 7

(2) 2 → 4 → 1 → 3 → 5 → 6 → 7 → 8

(3) 1 → 2 → 4 → 3 → 5 → 6 → 8 → 7

(4) 6 → 8 → 7 → 4 → 5 → 2 → 3 → 1

20) Choose the correct option from given below :-

| | **Innate immunity** | **Acquired immunity** |
|---|---|---|
| (1) | Skin, mucus coating | Antibody, antibiotics |

| | | |
|---|---|---|
| (2) | Acid in stomach, antiserum | Antibody, antivenum |
| (3) | Natural killer cell, interferon | Antiserum, antibiotics |
| (4) | Mucus coating, NK cell | Antivenom, antiserum |

(1) 1

(2) 2

(3) 3

(4) 4

21) Read the following statements (A-D)
(A) Health for a long time was considered as a state of body and mind where there was a balance of certain humors.
(B) Mind and mental state can affect our health
(C) Hippocrates disproved the 'good humor' hypothesis of health.
(D) It was thought that persons with blackbile belonged to hot personality and would have fevers.
How many of the above statements are true ?

(1) 4

(2) 1

(3) 2

(4) 3

22)

How many statement(s) are correct about acquired immunity ?
(A) Acquired immunity is a specific immunity.
(B) On second exposure of same antigen, this immunity shows anamnestic response.
(C) B & T Lymphocytes are responsible for acquired immunity

(D) It can differentiate between self and non self tissues.

(1) Two

(2) Three

(3) Four

(4) One

23) How many of the following factors are necessary for achieving good health ?
(a) awarness about disease
(b) Vaccination
(c) Proper disposal of wastes
(d) Control of disease vectors
(e) To Reduce infant and maternal mortality
(f) Economic prosperity

(1) a, b, d, e

(2) b, c, e, f

(3) a, b, c, d

(4) e, f only

24) Read the following statement about inflammation:
(a) Sequence of events are Rubor, Calor, Tumor, Dolar
(b) It acts as barrier towards microorganism
(c) It is specific barrier
(d) Temperature increased due to increased BMR
(e) Redness due to vasoconstriction
How many of above statement is/are correct?

(1) 4
(2) 3
(3) 2
(4) 1

25) Which of the following factors can affect our health : -
(A) Habits that we have or lack
(B) Infections
(C) Deficiencies with which a child is born
(D) Food and water we take
(E) Rest and exercise we give to our bodies
Choose the correct option : -

(1) Only A, B, C
(2) Only B, C
(3) A, B, C, D, E
(4) Only B, C, D, E

26) Infection of Ascaris usually occurs by

(1) Eating imperfectly cooked pork
(2) Tse - tse fly
(3) mosquito bite
(4) drinking water containing eggs of Ascaris

27) Find the correct match :-

(1) Epidermophyton - Filariasis
(2) Microsporum - Ring worm
(3) Shigella dysenteriae - Plague
(4) Haemophilus influenzae - Influenza

28) Which lymphoid organ act as a filter of blood by trapping blood borne microorganism ?

(1) Thymus
(2) Bone marrow
(3) Lymph node
(4) Spleen

29) N.K. cells make :-

(1) Physiological barrier of innate immunity
(2) Cellular barrier of acquired immunity
(3) Cytokine barrier of innate immunity
(4) Cellular barrier of innate immunity

30) When a quick immune response is required for infection of a deadly microbe, the patient is injected with :

(1) protein of pathogen
(2) inactivated or weakened pathogen
(3) preformed antibodies
(4) vaccine

31) Which of the following is trapped in the lymph nodes and responsible for the activation of lymphocytes present there and cause the immune response ?

(1) Antigen
(2) Antibody
(3) Pathogen
(4) Lymph fluid

32) In which one of the following options the two examples (given in column I) are correctly matched with their particular type of immunity (given in column-II) ?

| | **Column-I (Examples)** | **Column-II (Type of immunity)** |
|---|---|---|
| (1) | Saliva in mouth and tears in eyes | Physical barriers |
| (2) | Mucus coating of epithelium, urogential tract and HCl in stomach | Physiological barriers |
| (3) | Polymorphonuclear leucocytes and monocytes | Cellular barriers |
| (4) | Anti-tetanus and anti-snake bite injections | Active immunity |

(1) 1
(2) 2
(3) 3
(4) 4

33) Passive immunisation include :

(1) Introduction of antibodies directly
(2) Transfer of maternal Antibodies across placenta
(3) Transfer of Antigen directly
(4) Both 1 & 2

34) Which of the following sexual transmitting disease is curable if detected earlier ? **(Replica of Q.**

**98)**

(1) Hepatitis-B

(2) Genital herpes

(3) HIV

(4) Genital warts

35) Each molecule has four polypeptide chains ,two small called light chains and two longer called heavy chains (represented as $H_2L_2$). Which of the following is not directly related to above described molecule :-

(1) Army of proteins produced by B- lymphocytes

(2) Present in blood

(3) Humoral immunity

(4) Cell mediated immunity(CMI)

36) How many of of the following are non-phagocytic cells ?
(A) Monocytes
(B) Macrophage
(C) Neutrophils
(D) Natural killer cells

(1) One

(2) Two

(3) Three

(4) Four

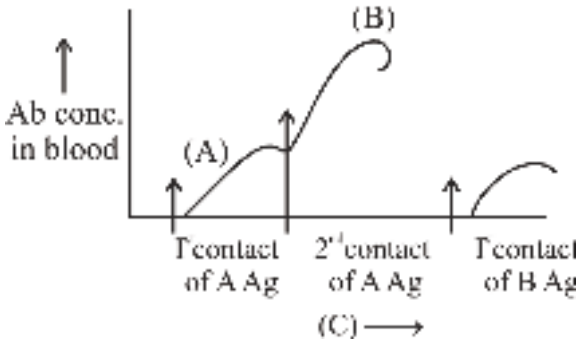
37) Choose incorrect statement for cell mediated immunity :-

(1) Mediated by T-lymphocytes

(2) Produce primary and secondary immune response

(3) Response involves production of antibodies

(4) Responsible for graft rejection.




38) Identify the correct A, B, C in given diagram :

(1) A-Primary IR, B-Secondary IR, C-Ab in blood

(2) A-Secondary IR, B-Primary IR, C-Time period

(3) A-Primary IR, B-Secondary IR, C-Time period

(4) A-Secondary IR, B-Primary IR, C-Ab in blood

39) Many diseases can be diagnosed by observing the symptoms in the patient. Which group of symptoms are indicative of pneumonia?

(1) Difficulty in respiration, fever, chills, cough, headache
(2) Constipation, abdominal pain, blood clots
(3) Nasal congestion and discharge, headache
(4) High fever, weakness, stomach pain, loss of appetite and constipation

40) Which of the following is/are **incorrect** about STD?

(1) Also called venereal diseases
(2) Early symptom includes, itching, fluid discharge, slight pain
(3) All STD are completely curable
(4) Can be prevented by use of condoms

41) Which of the following statements are not correct about venereal diseases ?

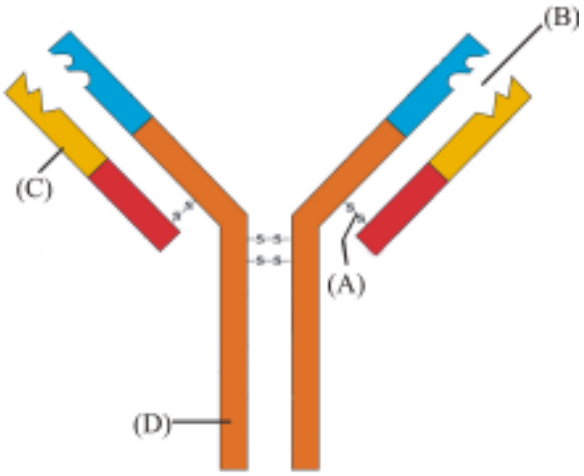
(1) Early symptoms of these are itching, fluid discharge, slight pain, swelling in the genital region
(2) They can lead to pelvic inflammatory diseases, abortions, cancer of reproductive tract.
(3) Genital herpes is one of the curable venereal disease.
(4) Genital warts caused by human papilloma and this virus can cause cervical cancer in females.

42) Identify the correct statements ?
(A) Saliva is considered to be a physiological barrier.
(B) $\alpha$-Interferons are virus specific.
(C) Our immune system feeds the memory after first encounter with pathogen.
(D) T-cells does not secrete antibodies but help B-cell to produce them.
(E) Antibodies are found in blood thats why their response is called humoral immune response.

(1) Four
(2) Five
(3) Three
(4) One

43) Which of the following is correct?

(1) B -cells provide CMI
(2) Interferons prevent the neighbouring cells from further viral infection
(3) As the age increases, thymosin increases
(4) T-cells produces antibodies




44) In the below diagram (A), (B), (C) and (D) are :-

(1) (A) – Light chain, (B) – Disulphide bonds,
(C) – Heavy chain, (D) – Paratope

(2) (A) – Disulphide bonds,
(B) – Antigen binding site,
(C) – Light chain, (D) – Heavy chain

(3) (A) – Heavy chain, (B) – Light chain,
(C) – Hydrogen bonds, (D) – Paratope

(4) (A) – Disulphide bonds,
(B) – Antigen presenting site,
(C) – Heavy chain, (D) – Light chain

45) The following two organs are example of secondary lymphoid organs :-

(1) spleen and thymus

(2) lymph nodes and thymus

(3) spleen and GALT

(4) bone marrow and MALT

46) Match the column I with column II and choose the correct option.

| | **Column I** | | **Column II** |
|---|---|---|---|
| **a.** | Incomplete dominance | **1.** | ABO blood grouping |
| **b.** | Co-dominance | **2.** | Snapdragon |
| **c.** | Multiple allele | **3.** | Garden pea |
| **d.** | Dominance | **4.** | Fruit-files |

(1) a → 2, b → 4, c → 3, d → 1

(2) a →1, b → 2, c → 4, d → 3

(3) a → 3, b → 4, c → 1, d → 2

(4) a → 2, b → 1, c → 1, d → 3

47) **Assertion :** Dominant feature of a character can be expressed in homozygous as well as heterozygous.

**Reason :** Dominance and Recessiveness are two autonomous features of a gene always.

(1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) If Assertion is true statement but Reason is false

(4) If both Assertion and Reason are false statements

48) Codominance is different from incomplete dominance as in codominance

(1) The hybrid is intermediate

(2) Dominant is expressed in $F_1$

(3) Both the alleles are expressed equally

(4) Genotype ratio is 1 : 1

49) Match column I with column II and choose the correct option:

| | **Column i** | | **Column II** |
|---|---|---|---|
| **a.** | Reginald C. Punnett | **1.** | Sex determination |
| **b.** | Mendel | **2.** | Genotype probability |
| **c.** | Sutton and Boveri | **3.** | Linkage and recombination |
| **d.** | T.H. Morgan | **4.** | Laws of inheritance |
| **e.** | Henking | **5.** | Parallel behaviour of genes and chromosomes |
| | | | |

(1) a - 2, b - 4, c - 3, d - 1, e - 5

(2) a - 4, b - 5, c - 1, d - 2, e - 3

(3) a - 2, b - 4, c - 5, d - 3, e - 1

(4) a - 5, b - 3, c - 2, d - 1, e - 4

50) In the $F_2$ generation of a Mendelian dihybrid cross, the number of phenotypes and genotypes are

(1) Phenotypes - 4; genotypes - 16

(2) Phenotypes - 9; genotypes - 4

(3) Phenotypes - 4; genotypes - 8

(4) Phenotypes - 4; genotypes - 9

51) All genes located on the same chromosomes

(1) Form different groups depending upon their relative distance

(2) Form one linkage group

(3) Will not form any linkage groups

(4) Form interactive groups that affect the phenotype

52)

Read the following statements about polygenic inheritance and choose the correct option.
I. In a polygenic trait, the phenotype reflects the contribution of each allele, i.e., the effect of each allele is additive.
II. Besides the involvement of multiple genes, polygenic inheritance also takes into account the influence of environment.
III. The number of each type of alleles in the genotype would determine the darkness or lightness of the skin in an individual.
IV. The dominant forms A, B, and C responsible for the dark skin colour and the recessive forms a, b, and c for light skin colour.
Incorrect statement(s) is/are

(1) Only I

(2) Only I and III

(3) Only II and IV

(4) None of the above

53) If skin colour in humans is controlled by three gene pairs, then what is the number of intermediates and parental combination in $F_2$, generation?

(1) 2, 62

(2) 62, 2

(3) 2, 14

(4) 14, 2

54) Given below are some statements:-
(i) Occur in pairs
(ii) Segregate at the time of gamete formation such that only one of each pair is transmitted to a gamete.
(iii) Independent pairs segregate independly of each other.
The above statements are valid for:-

(1) Genes only

(2) Chromosomes only

(3) Both genes & chromosomes

(4) Neither genes nor chromosomes

55) **Statement I:** Traits which are generally controlled by three or more genes are called pleiotropic traits.
**Statement II:** Henking gave a name X body for a chromosome in insects, and also explained its significance.

(1) Both statement I and statement II are true

(2) Both statement I and statement II are false

(3) Statement I is true, and statement II is false

(4) Statement I is false, and statement II is true.

56) Sex determination in grasshopper, humans, and Drosophila is similar because

(1) Females are hemizygous

(2) Male have one X-chromosome and female have two X-chromosomes

(3) Females shows heterogamety

(4) The ratio of autosomes and sex chromosomes is the same in all three organisms

57) Match the column I with column II and choose the correct combination from the options given below.

| | **Column I** | | **Column II** |
|---|---|---|---|
| **a.** | Chromosomal disorder | **1.** | Drosophila |
| **b.** | XO type sex determination | **2.** | Grasshopper |
| **c.** | XY type sex determination | **3.** | Fowls |
| **d.** | ZW type sex determination | **4.** | Klinefelter's syndrome |

(1) a → 1, b → 2, c → 3, d → 4

(2) a → 4, b → 3, c → 2, d → 1

(3) a → 4, b → 2, c → 1, d → 3

(4) a → 2, b → 1, c → 2, d → 3

58) An individual with AaBb genotype is producing four types of gametes as AB, aB, Ab and ab by meiosis. If the frequency of these gametes is 25% each then, it explains

(1) Incomplete linkage

(2) Mutations

(3) Independent assortment

(4) Complete linkage

59) Total number of phenotypic and genotypic classes possible from selfing of a plant with genotype aaBbCCDdEEFf are respectively

(1) 64, 729

(2) 8, 27

(3) 32, 243

(4) 27, 35

60) Select incorrect statement w.r.t. polygenic inheritance

(1) It controls quantitative traits

(2) Parental population remains constant irrespective of polygenes number

(3) A single dominant allele expresses only a unit of the trait

(4) One of the parental phenotype is expressed in $F_1$ individuals

61) How many dihybrid crosses are possible from the contrasting traits studied by Mendel in pea?

(1) 21

(2) 42

(3) 14

(4) 7

62) In a dihybrid cross , if first gene shows dominance and second gene shows incomplete dominance. Then, what will be the phenotypic ratio in $F_2$ generation.

(1) 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1

(2) 9 : 3 : 3 : 1

(3) 1 : 1 : 2 :3 : 3 : 6

(4) 1 : 1 : 1 : 1 :1 : 1

63) What would be the number of tall and pink plants, if tall and pink plants (TtRr) are selfed? Given that first character shows conplete dominance and second character shows incomplete dominance

(1) 9/16

(2) 8/16

(3) 6/16

(4) 3/16

64) What will be the sum total of genotypic classes and phenotypic classes in a trihybrid polygenic cross in $F_2$ generation?

(1) 16

(2) 35

(3) 34

(4) 9

65) A dihybrid test cross produced the following offsprings
AB/ab 960
Ab/ab 60
aB/ab 40
ab/ab 940
What is the distance between gene A and B?

(1) 5 map units

(2) 100 map units

(3) 20 map units

(4) 10 map units

66)

In Morgan's experiment on Drosophila types of gametes produced by wild male and wild female respectively:-

(1) 2, 2

(2) 2, 4

(3) 4, 4

(4) 4, 2

67) What is the sequence of genes on chromosomes if frequency of crossing over between A -B is 20%, A-C is 15%, B-C is 5%, A-D is 10% and B-D is 30%?

(1) A B C D

(2) C B D A

(3) D A C B

(4) D A B C

68) The phenotypic expression 3 : 1 obtained at the $F_2$ generation in monohybrid cross is best explained by

(1) Law of dominance

(2) Law of segregation

(3) Law of independent assortment

(4) All of the above

69) Which among the following was used as a starting point in the case of human genome sequencing project?

(1) Frequency of linkage

(2) Linkage map

(3) Genes showing independent assortment

(4) Total polygenes in human

70) If two genes a and b are linked and shows 20% recombination. The proportion of gametes produced in $F_1$ by a dihybrid ++/ab derived from a cross between ++/++ and ab/ab would be

(1) ++ 80% : ab 20%

(2) ++ 50% : ab 50%

(3) ++ 40% : ab 40% : a 10% : + b 10%

(4) ++ 20% : ab 20% : a 20% : + b 20%

71) Consider the cross PpQQRrSs × PpqqRRSs. Find out the proportion of progeny with genotype ppQqRRss in the resulting population.

(1) $\frac{1}{64}$

(2) $\frac{1}{32}$

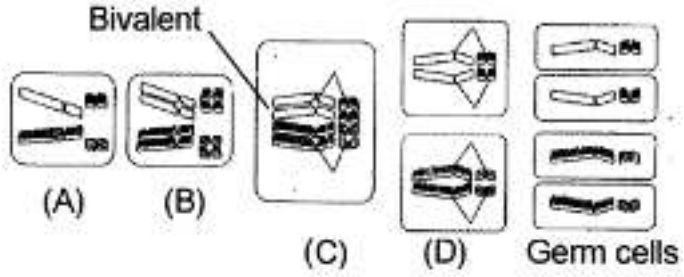
(3) $\frac{1}{16}$

(4) $\frac{1}{128}$

72) The phenotype of an individual may be affected if the modified allele produces
A. The normal/less efficient enzyme
B. No enzyme at all
C. A non-functional enzyme

(1) Only A is correct
(2) C & A are correct
(3) Only B is correct
(4) B & C are correct

73) The figure below shows the chromosomes segregation during germ cells formation with four stages labelled as (A), (B), (C) and (D). Select the right option giving all the four stages correctly identified




(1) (A) Interphase (C) Telophase-l
(B) Metaphase-l (D) Anaphase-ll

(2) (A) $G_1$ (B) $G_2$
(C) Anaphase-l (D) Anaphase-ll

(3) (A) $G_1$ (B) $G_2$
(C) Metaphase-l (D) Anaphase-I

(4) (A) $G_1$ (B) $G_2$
(C) Anaphase-l (D) Metaphase-II

74) **Assertion :** Height expression in human beings is a polygenic inheritance.
**Reason :** Genotype with two dominant alleles and three recessive alleles will have intermediate skin colour.

(1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) If Assertion is true statement but Reason is false
(4) If both Assertion and Reason are false statements

75) **Assertion :** Sickle cell anaemia is resulted due to a transversion.
**Reason :** A pyrimidine 'T' is substituted by 'A' in DNA sequence normally coding for glutamic acid.

(1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) If Assertion is true statement but Reason is false
(4) If both Assertion and Reason are false statements

76) A cross was made between two individuals. Some offsprings obtained were parental and some were recombinants. The maximum percentage of recombinants and parental types in any cross can be respectively

(1) 50% 100%
(2) 100% 50%
(3) 70% 70%
(4) 80% 80%

77) When a cross was made between pink flowered and white flowered snapdragon plants, what will be the percentage of white flowered plants in progeny?

(1) 0%
(2) 25%
(3) 50%
(4) 100%

78) Females and males are heterogametic in ____(i)____ and _____(ii)____ respectively

(1) (i) Grasshopper, (ii) Birds
(2) (i) Birds, (ii) Grasshopper
(3) (i) *Drosophila,* (ii) Butterflies
(4) (i) Moth, (ii) Birds

79) Select the **correct** statements out of the following.
A: The sex determination in honey bee is based on the number of sets of chromosomes an individual receives.
B: In human beings, Y chromosome carries Sry gene.
C: In human males, all sperms contain same number and type of autosomes and sex chromosome.

(1) A and B
(2) B and C
(3) Only C
(4) C and A

80) Which among the following can be studied in a population?

(1) AB blood group
(2) MN blood group
(3) Pleiotropy
(4) Multiple alleles

81) A pea plant is bearing all white flowers. Which of the following statements cannot be correct about this plant?

(1) This plant is homozygous for flower colour
(2) All the seeds produced by this plant will always be wrinkled

(3) The flowers in this plant may be axial

(4) This plant may be heterozygous for pod colour

82) Heterozygous tall and violet flowered pea plants were selfed and total 512 seeds are collected. What will be total number of seeds for both heterozygous traits?

(1) 128

(2) 256

(3) 384

(4) 64

83) Mark the odd one out (w.r.t. $F_2$ generation of Mendelian dihybrid cross)

(1) Frequency of TtRR genotype = 12.5%

(2) Frequency of ttrr genotype = 6.25%

(3) Frequency of TTRR genotype = 6.25%

(4) Frequency of ttRr genotype = 25%

84) Find out the frequency of AabbCcDdee if parents are AabbCCddEe and AabbccDdee

(1) 0.78%

(2) 12.5%

(3) 25%

(4) 50%

85) Holandric genes are present on

(1) X-chromosomes

(2) Y- chromosomes

(3) Sex-chromosomes as well as autosomes

(4) Autosomes

86) Cytoplasmic male sterility in maize is due to defective

(1) Mitochondria

(2) Lysosome

(3) Golgi body

(4) Leucoplast

87) In Antirrhinum (Snapdragon), a red flower was crossed with a white flower and in F, generation pink flowers were obtained. When pink flowers were selfed, the F, generation showed white, red and pink flowers. Choose the incorrect statement from the following:

(1) This experiment does not follow the Principle of Dominance.

(2) Pink colour in $F_1$ is due to incomplete dominance.

(3) Ratio of $F_2$ is $\frac{1}{4}$(Red): $\frac{2}{4}$(Pink): $\frac{1}{4}$(White)

(4) Law of Segregation does not apply in this experiment

88) Which of the following pairs is wrongly matched:-

(1) Starch synthesis in pea : Multiple alleles

(2) ABO blood grouping : Co-dominance

(3) T.H. Morgan : Linkage

(4) XO type sex determination : Grasshopper

89)

Based upon results of dihybrid cross performed by Mendel, total no. of offsprings dominant as well as recessive for atleast one character were :-

(1) 9, 1

(2) 15, 7

(3) 9, 7

(4) 15,1

90) Multiple alleles are present

(1) On non/sister chromatids

(2) On different chromosomes

(3) At different loci on the same chromosomes

(4) At the same locus of the chromosomes

ANSWER KEYS

## PHYSICS

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 2 | 3 | 2 | 3 | 4 | 2 | 1 | 3 | 2 | 1 | 1 | 2 | 1 | 1 | 3 | 4 | 1 | 3 | 1 | 2 |
| Q. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| A. | 3 | 1 | 3 | 2 | 3 | 4 | 3 | 3 | 1 | 2 | 3 | 2 | 4 | 2 | 4 | 2 | 4 | 3 | 2 | 3 |
| Q. | 41 | 42 | 43 | 44 | 45 | | | | | | | | | | | | | | | |
| A. | 4 | 1 | 2 | 3 | 1 | | | | | | | | | | | | | | | |

## CHEMISTRY

| Q. | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 2 | 3 | 3 | (2) | 2 | 4 | 1 | 1 | 3 | 2 | 3 | 2 | 1 | 1 | 4 | 3 | 1 | 1 | 1 | 4 |
| Q. | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |
| A. | 1 | 2 | 1 | 4 | 2 | 2 | 2 | 4 | 4 | 2 | 3 | 2 | 3 | 4 | 3 | 4 | 4 | 2 | 4 | 3 |
| Q. | 86 | 87 | 88 | 89 | 90 | | | | | | | | | | | | | | | |
| A. | 4 | 4 | 2 | 4 | 3 | | | | | | | | | | | | | | | |

## BIOLOGY

| Q. | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 2 | 4 | 4 | 1 | 3 | 4 | 3 | 3 | 2 | 2 | 4 | 3 | 3 | 1 | 1 | 3 | 2 | 2 | 1 | 4 |
| Q. | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 |
| A. | 1 | 3 | 3 | 2 | 3 | 4 | 2 | 4 | 4 | 3 | 1 | 3 | 4 | 4 | 4 | 1 | 3 | 3 | 1 | 3 |
| Q. | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 |
| A. | 3 | 1 | 2 | 2 | 3 | 4 | 3 | 3 | 3 | 4 | 2 | 4 | 2 | 3 | 2 | 2 | 3 | 3 | 2 | 4 |
| Q. | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 |
| A. | 1 | 3 | 3 | 3 | 1 | 2 | 3 | 1 | 2 | 3 | 2 | 4 | 2 | 3 | 1 | 1 | 3 | 2 | 1 | 4 |
| Q. | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 | | | | | | | | | | |
| A. | 2 | 1 | 4 | 2 | 2 | 1 | 4 | 1 | 2 | 4 | | | | | | | | | | |

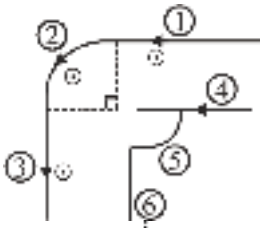
SOLUTIONS

PHYSICS

1) We shall use $\oint \vec{B}\vec{dl} = \mu_0 I$ where I is the current enclosed by loop
Net current enclosed by path a is zero.
Net current enclosed by path c is 1A.
Net current enclosed by path d is 3A.
Net current enclosed by path b is 5A.

2) $\odot B_1 = \frac{\mu_0 i}{4\pi R}, \overset{\odot}{B_2} = \frac{\mu_0 i}{8R}$

$\overset{\odot}{B_3} = \frac{\mu_0 i}{4\pi R}$, $B_4 = 0$, $B_6 = 0$




$\otimes B_5 = \frac{\mu_0 i}{8R}$

$B_{net} = \frac{\mu_0 i}{2\pi R}$

3) $\sin\theta = \frac{dqB}{\sqrt{2m\,K.E}}$

$\sin\theta = \frac{10^{-1} \times 1.6 \times 10^{-19} \times 0.51}{\sqrt{2 \times 1.6 \times 10^{-27} \times 500 \times 1.6 \times 10^{-19} \times 10^3}}$

$\sin\theta = \frac{1}{2} \Rightarrow \quad \theta = 30°$

4) $\vec{F} = q[\vec{E} + (\vec{V} \times \vec{B})]$

5) $B = \frac{\mu_0 i}{2r} - \frac{\mu_0 i}{2(2r)} + \frac{\mu_0 i}{2(2^3 r)} + \ldots\infty$

$B = \frac{\mu_0 i}{2r}\left[1 - \frac{1}{2} + \frac{1}{2^2} - \frac{1}{2^3} + \ldots\infty\right]$

$B \simeq \frac{\mu_0 i}{2r}\left[\frac{1}{1-\left(-\frac{1}{2}\right)}\right]$

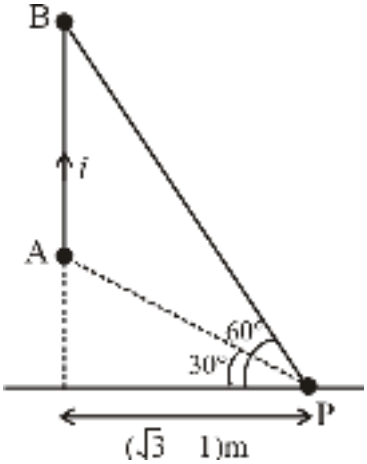
$$B = \frac{\mu_0 i}{3r}$$

6)

Sol : $\frac{\mu_0 i}{3\pi a}\sqrt{4+\pi^2}$

7) $B = \frac{\mu_0 i}{4\pi d}(\sin\theta_1 + \sin\theta_2)$

$= \frac{10^{-7}\times 100}{\sqrt{3}-1}\left[\frac{\sqrt{3}}{2}-\frac{1}{2}\right] = 5\times 10^{-6}\,T$




8) $B_x = \frac{\mu_0 i r^2}{2\left[r^2+\left(\frac{d}{2}\right)^2\right]^{\frac{3}{2}}} \quad \frac{\mu_0 i r^2 \times 8}{2[4r^2+d^2]^{\frac{3}{2}}}$

$B_y = \frac{\mu_0 i(r^2)}{2[4r^2+d^2]^{\frac{3}{2}}} \quad \frac{\mu_0 i r^2 \times 4}{2[4r^2+d^2]^{\frac{3}{2}}}$

$\frac{B_y}{B_x} = \frac{1}{2}$

9)

Solution :- $\frac{\mu_0 I}{4\pi a}\left(1-\frac{b}{\sqrt{a^2+b^2}}\right)$

10)

Sol : - $\theta = 2$ rad

11)

Sol:- $1.14\times10^{-5}T$

12)

Sol:- $\left(\frac{\pi}{2}+1\right)a^2I\hat{k}$

13)

Four times

14)

A charge particle cannot be accelerated by uniform magnetic field

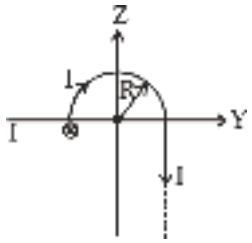
15)

Sol:- $2iBL$

16)

Sol:- $d\vec{B}=\frac{\mu_0}{4\pi}i\left(\frac{d\vec{l}\times\vec{r}}{r^3}\right)$

17)




Magnetic field at 'O'

$$\vec{B}_1=\frac{\mu_0 I}{4\pi R}\left(-\hat{k}\right)$$

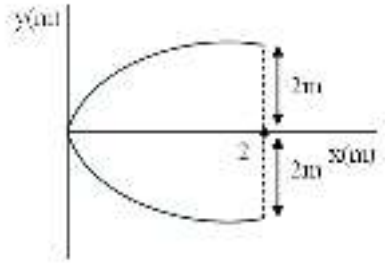
$$\vec{B}_2=\frac{\mu_0 I}{4R}\left(-\hat{i}\right)$$

$$\vec{B}_3=\frac{\mu_0 I}{4\pi R}\left(-\hat{i}\right)$$

$$\vec{B}_{at\,'O'}=\frac{\mu_0 I}{4R}\left(1+\frac{1}{\pi}\right)\left(-\hat{i}\right)+\frac{\mu_0 I}{4\pi R}\left(-\hat{k}\right)$$

18) **1. Explanation:** A Conducting wire bent in the form of a parabda $y^2 = 2x$ carries a Current i=2A as show in figure. This wire placed in uniform magnetic field B = -4k. Hind the magnetic

farce on wire.
**2. Concept:** Magnetic farce on current carrying wire.
**3 Formula:** F = iBL
**4. Mathematical Calculation:**



at x = 2
$y^2 = 2x$
$y = \sqrt{2x^2}$
y = 2
the displacement length of wire L = 4m
$F = i(\vec{L} \times \vec{B})$
$F = 2(4\hat{i} \times 4\hat{k})$
$F = -32\hat{i}$
5. Correct answer with option: (3) $[-32\hat{i}]$
6. Estimated time : 2-3 minutes
**7. Level: Medium**

19) $B = \frac{\mu_0 i}{4\pi r} \times \alpha$
$\frac{B_1}{B_2} = \frac{\frac{\mu_0 i_1}{4\pi r}\alpha_1}{\frac{\mu_0 i_2}{4\pi r}\alpha_2} = \frac{i_1\alpha_1}{i_2\alpha_2}$

20) According to question resistance of wire ADC is twice that of wire ABC, Hence current flows through ADC i.e. $\frac{i_2}{i_1} = \frac{1}{2}$ Also $i_1 + i_2 = i$
$\Rightarrow i_1 = \frac{2i}{3}$ and $i_2 = \frac{i}{3}$
Magnetic field at centre O due to wire AB and BC (part 1 and 2) $B_1 = B_2 = \frac{\mu_0}{4\pi}.\frac{2i_1 \sin 45^\circ}{a/2}\otimes$
$= \frac{\mu_0}{4\pi}.\frac{2\sqrt{2}i_1}{a}\otimes$
and magnetic field at centre O due to wires AD and DC (i.e. part 3 and 4)
$B_3 = B_4 = \frac{\mu_0}{4\pi}.\frac{2\sqrt{2}i_2}{a}\odot$
also $i_1 = 2i_2$. So $(B_1 + B_2) > (B_3 + B_4)$
Hence net magnetic field at centre O
$B_{net} = (B_1 + B_2) - (B_3 + B_4)$
$= 2 \times \frac{\mu_0}{4\pi}.\frac{2\sqrt{2} \times \left(\frac{2}{3}i\right)}{a} - \frac{\mu_0}{4\pi}.\frac{2\sqrt{2}\left(\frac{1}{3}\right) \times 2}{a}$

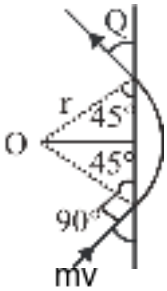
$= \frac{\mu_0}{4\pi}.\frac{4\sqrt{2}I}{3a}(2-1)\otimes = \frac{\sqrt{2}\mu_0 i}{3\pi a}\otimes$

21) $\oint \vec{B}.\vec{d\ell} = \mu^0 i$

B.$2\pi\left(\frac{3R}{2}\right) = \frac{\mu_0 I \times \left[\pi\left(\frac{3}{2}R\right)^2 - \pi R^2\right]}{\pi\left[(2R)^2 - R^2\right]}$ B.$3\pi R = \frac{\mu_0 I}{3} \times \frac{5}{4}$ ; $B = \frac{5\mu_0 I}{36\pi R}$

22) $\frac{PQ}{2}$ = r cos 45°

PQ = 2r cos 45°




r = $\frac{mv}{qB}$

23)

Sol : $B_0IL$

24)

Sol : $\frac{\mu_0 i}{2\pi l}$

25)

Sol:- $\sqrt{\frac{2Vm}{q}}\left(\frac{1}{B}\right)$

26)

Zero

27)

$E \neq 0$, $B = 0$

28)

Sol :- $\left(\frac{1}{2}\right)^{3/2}$

29)

Sol : $\frac{\mu_0 I}{8}\left(\frac{R_1+R_2}{R_1R_2}\right)$

30)

2 cm

31) By using,

$\oint \vec{B}\cdot\vec{d\ell} = \mu_0 \Sigma I$ [Ampere's law]

For loop A : $\oint \vec{B}\cdot\vec{d\ell} = \mu_0(0) = 0$

For loop B : $\oint \vec{B}\cdot\vec{d\ell} = \mu_0 I$

For loop C : $\oint \vec{B}\cdot\vec{d\ell} = -\mu_0 I$

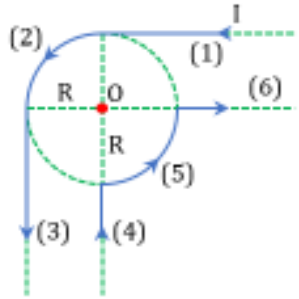
For loop D : $\oint \vec{B}\cdot\vec{d\ell} = -\mu_0 I$

32) (b) Point 'O' is located on the extended axis of straight parts (4) and (6), so $B_4 = B_6 = 0$

Magnitude and direction of magnetic field at 'O' due to wire (1) and (3) is same so

$B_1 = \frac{\mu_0 I}{4\pi R}(\sin 0° + \sin 90°) = \frac{\mu_0 I}{4\pi R} = B_3$

Due to circular arc (2) and (5) magnetic field at 'O' circular arc (2) and (5) is same so :-




$B_2 = \frac{\mu_0 I(\pi/2)}{4\pi R} = \frac{\mu_0 I}{8R} = B_5$

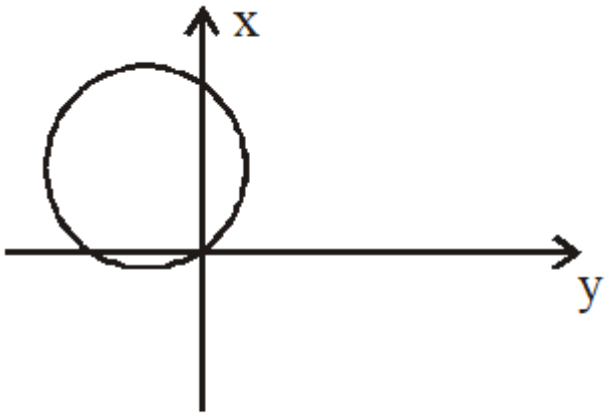
so resultant magnetic field at 'O' :-

$B_0 = B_1 + B_3 + B_2 + B_5$

$= \frac{\mu_0 I}{4\pi R} + \frac{\mu_0 I}{4\pi R} + \frac{\mu_0 I}{8R} + \frac{\mu_0 I}{8R}$

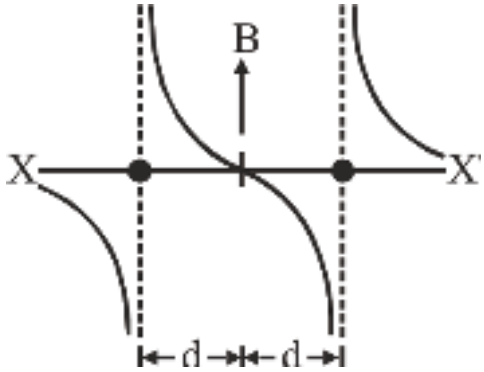
$= \frac{\mu_0 I}{2\pi R} + \frac{\mu_0 I}{4R} = \frac{\mu_0 I}{4R}\left(\frac{2}{\pi} + 1\right)$

33)




circle in xy plane

Sol :




34)

35)

Sol : $\frac{\mu_0 i}{2\pi\left(x^2+y^2\right)}\left(x\hat{j}-y\hat{i}\right)$

36)

Sol : $-\left(\frac{\mu_0 i}{4R}\right)\hat{k}$

37)

Sol : Zero even if the current 'i' leaves point D instead of point E

38) $\frac{\mu_0 I N}{2(b\quad a)}\frac{\log_e(b)}{a}$

39)

Sol : $4\pi\varepsilon_0\left[\frac{ab}{b-a}+\frac{c^2}{c-b}\right]$

40)

$C_m = KC_{air}$

$E_m = \frac{E_{air}}{K}$

$V_m = \frac{V_{air}}{K}$

41) for equal distance division :-

$\frac{2}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2}$

$\Rightarrow \frac{2}{k_{eff}} = \frac{k_1+k_2}{k_1k_2} \Rightarrow k_{eff} = \frac{2k_1k_2}{k_1+k_2}$

now this effective dielectric has equal area division with $k_3$

$\Rightarrow k_{final} = \frac{k_{eff}+k_3}{2}$

$= \frac{\frac{2k_1k_2}{k_1+k_2}+k_3}{2} = \left(\frac{k_1k_2}{k_1+k_2}+\frac{k_3}{2}\right)$

$\Rightarrow C_{final} = k_{final}C$

$\Rightarrow$ C = option (4)

42)

Electric field between plates of capacitor is given by,

$E = \frac{\sigma}{k\in_0}$

(K → Dielectric constant)

43)

Sol : $\frac{4\in_r}{3\in_r+1}$

44)

Sol : $\frac{\varepsilon_0 A}{d-t\left(1-\frac{1}{k}\right)}$

45) $\frac{dv}{dx} = -E$, Electric field in a dielectric falls to $\frac{E_0}{k}$, where '$E^0$' is electric field in free space.

46)

In x $\Rightarrow O_2^{-2}$ is present
In y $\Rightarrow O_2^{-1}$ is present
B.E. < B.O.
$\therefore$ BE $O_2^- > O_2^{2-}$
y > x

47)

A = $[Co(H_2O)_6]Cl_3$ → Max Electrical Conductance
B = $[Co(H_2O)_5Cl]Cl_2.H_2O$
C = $[Co(H_2O)_4Cl_2]Cl.2H_2O$ → Max nonionisable $Cl^-$
$[Co(H_2O)_3Cl_3].3H_2O$ ⇒ Not possible

48) No. of G-I = 2
No.of Linkage = 3
Total = 2 × 3 =6

49) is - platin
⇒ $[Pt(NH_3)_2Cl_2]$
Prussian Blue
$Fe_4[Fe(CN)_6]_3$

50)

**Explanation:**
The question asks you to identify the incorrect stability order among the given sets of coordination complexes.

**Concept: (Stability chelate effect, ligand strength, Metal ion size, Metal oxidation state of coordination compounds)**

A. **Chelate effect:** Polydentate ligands (en, trien) create more stable complexes. (Correct)
B. **Ligand strength:** Stronger field ligands ($NO_2^-$) create more stable complexes than weaker ones ($NH_3$, $H_2O$). (Incorrect)
C. **Metal ion size:** Stability increases down a group for similar complexes.
D. **Metal oxidation state:** Higher oxidation states generally increase complex stability.

**Final Answer:** Option (2)

51)

**Explanation –** The question asks which of the given options results in a blue coloration.

**Concept – (Various reaction & the colour product formed)**

$CuSO_4(aq) + 2NH_4OH(aq) \rightarrow Cu(OH)_2(s) + (NH4)_2SO_4(aq)$
pale blue

$4FeCl_3 + 3K_4[Fe(CN)_6] \rightarrow Fe_4[Fe(CN)_6] + 12KCl$

deep blue pigment
$CuSO_4(s) + 6H2O(l) \rightarrow [Cu(H_2O)_6]SO_4(aq)$
light blue color

**Final Answer – Option (4)**

52) $K = \frac{1}{\beta_4}$

$= \frac{1}{2.1 \times 10^{13}}$

$= 4.7 \times 10^{-14}$

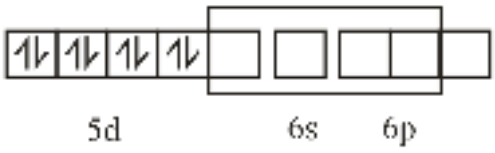
53) due to $\Delta_t <$ P.E.

54) **Explaining the question :** Identify which of the listed cobalt complexes are chiral and thus exhibit optical isomerism.
**Concept :** Optical Isomerism **Solution :** Cis-$[Co(en)_2Cl_2]^+$ is chiral due to the adjacent arrangement of ligands lacking a plane of symmetry. $[Co(en)_3]^{3+}$ is also chiral as the tris-chelated structure is non-superimposable on its mirror image.

55)

As per concept

56) $[PtCl_4]^{-2}$

$Pt^{+2} = 5d^8 =$ 


57) M–C B.O./C–O B.L. $\propto$ –ve charge

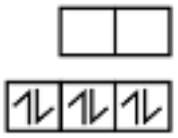
58)

$[Ni(CN)_5]^{3-}$ square pyramidal complex

59) $Zn^{+4}$ & $Hf^{+4}$ has same ionic size because of lanthanoid contraction.

60) $[Co(en)_2Br_2]Cl$ will show 6 isomers.

61) (1) In this compound $Fe^{+2} \rightarrow 3d^6$



$d^2sp^3$ hyb.

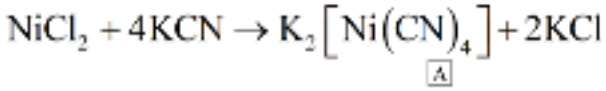
but this compound is paramagnetic due to presence of superoxide as a ligand.
(2) $[Ni(dmg)_2]$ is more stable due to more chelate ring.
(3) EAN of $[V(CO)_6]$
$= 23 - 0 + 2 \times 6$
$= 23 + 12 = 35$
EAN $[V(CO)_6]^-$
$23-(-1) + 2 \times 6$
$24 + 12 = 36$
After gaining $1e^-$ $[V(CO)_6]$ reduces to $[V(CO)_6]^-$ as it starts following EAN Rule
(4) CO, $CN^-$, $NO^+$ are σ-donor & p acceptor.

62)

Formula of brown ring complex is
$[Fe(H_2O)_5NO]SO_4$

63) Chelating agents form more stable complexes thus, these are used to remove toxic metal ions.

64)



$NiCl_2 + 4KCN \rightarrow \underset{A}{K_2[Ni(CN)_4]} + 2KCl$

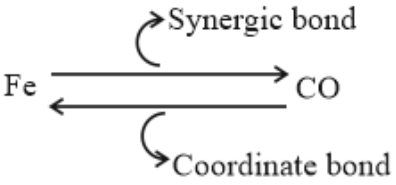
Pot. tetracyanonickelate (II)



$\underset{A}{K_2[Ni(CN)_4]} + \underset{(conc)}{4HCl} \rightarrow \underset{B}{K_2[NiCl_4]}$

Pot. tetrachloridonickelate (II)

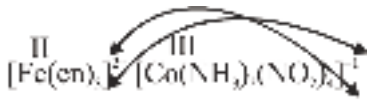
65)




CO is $\pi$ - acidic ligand

66) As +ve charge on complex M - C bond strength decreases
C - O bond strength increases.

67) $[Pt(H_2O)_6]^{+4}$ $\Delta_0 > P$
with 4d/5d metal all ligand acts as SFL



68) $[Fe(en)_3]$ $[Co(NH_3)_2(NO_2)_4]$

69)

C-C bond length
$C_2H_4 < K[PtCl_3(C_2H_4)]$

70)

CFSE $\Rightarrow -0.4\Delta_0 \times 1 = -0.4\ \Delta_0$
$\Rightarrow -0.4 \times 242 = -96.80$ kJ/mol

71) $[Fe(en)_3]^{2+}$ is obtained as X
Since 'en' is a strong field ligand, so the complex will be low spin & diamagnetic
Hybridization will be $d^2sp^3$ (i.e inner orbital complex)

72)

$[Ni(dmg)_2]$ having 4 rings

73)

$Pt^{+4} \rightarrow$ C.N = 6 ; $Pt^{+2} \rightarrow$ CN = 2
(1) $[Pt(H_2O)_6]Cl_4$ 100% chloride ionisable
(2) $[Pt(H_2O)_4Cl]Cl_3$ 75% chloride ionisable
(3) $[Pt(H_2O)_2Cl_2]$ 0% chloride ionisable
(4) $[Pt(H_2O)_3Cl_3]Cl$ 25% chloride ionisable

74)

**Explanation -** Identify the compound that violates the 18-electron rule when CO ligands are substituted by NO in neutral metal carbonyls.
**Concept - (EAN)**
**Electron Counting:** CO contributes 2 electrons, and NO contributes 3 (linear).
**18-Electron Rule:** Stable transition metal complexes usually follow the 18-electron rule.
**Ni Violation:** $Ni(CO)_2(NO)_2$ results in 20 electrons, violating the 18-electron rule.
**Correct Formulas:** The other options ($Cr(CO)_3(NO)_2$, $Fe(CO)_2(NO)_2$, $Cr(NO)_4$) all comply with the 18-electron rule.
**Final Answer - Option (4)**

75)

Formation of 0.2 mol of AgCl from 0.1 mol of the complex shows that there are two ionizable

Cl-atoms. Formula of the complex is $[Co(NH_3)_5Cl]Cl_2$, i.e., 1 : 2 type electrolyte.
$[Co(NH_3)_5Cl]Cl_2 \rightleftharpoons [Co(NH_3)_5Cl]^{2+} + 2Cl^-$

76) $AlCl_3 + 6H_2O \rightarrow \underset{sp^3d^2}{[Al(H_2O)_6]^{+3}_{(aq)}} + 3\ Cl^-(eq)$

77) $NO_2$ is brown coloured gas while in dimer it is colourless

78)

**Generated by Allie**

**Problem Statement:** The question is asking which metal from the given options (Mn, Tl, Sc, In) can replace or displace aluminum (Al) from its compound called alum. In other words, it wants to know which metal is more reactive than aluminum and can take aluminum's place in alum.

**Underlying Concept:** This problem is based on the concept of displacement reactions and activity series of metals. A more reactive metal can displace a less reactive metal from its compounds. The reactivity is related to the ability to lose electrons, which can be analyzed using standard reduction potentials (EÂ°). A metal with a more negative standard reduction potential is more reactive and can reduce the ion of a less reactive metal, displacing it from the compound.

**Step-by-Step Calculation:** We compare the standard reduction potentials of the metals involved: 1. Scandium reaction: $Sc^{3+} + 3e^- \rightarrow Sc \quad E^\circ = -2.09\,V$ 2. Aluminum reaction: $Al^{3+} + 3e^- \rightarrow Al \quad E^\circ = -1.66\,V$ Since Sc has a more negative EÂ° value (-2.09 V) compared to Al (-1.66 V), Sc acts as a stronger reducing agent and will displace Al from its compound.

**Tips and Tricks:** Remember that a metal's ability to displace another from its compound depends on its position in the reactivity series, which correlates inversely with its standard reduction potential.

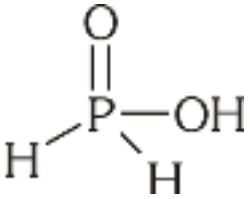
**Common Mistakes:** Assuming that all listed metals can displace aluminum without checking their reduction potentials, or confusing oxidation with reduction potentials.

**Why Other Options Are Incorrect?:** Mn, Tl, and In have less negative or higher standard reduction potentials than Al, so they are less reactive and cannot displace Al from alum.

79) Theory Based
$Me_3SiCl \rightarrow$ forms dimer only

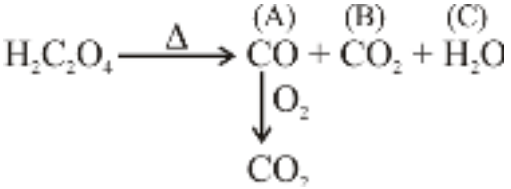
80) Strong reducing behaviour of $H_3PO_2$

All oxy-acid of phosphorus which contain
P–H bond act as reductant.

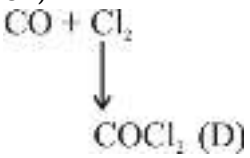


presence of one –OH group and two P–H bonds

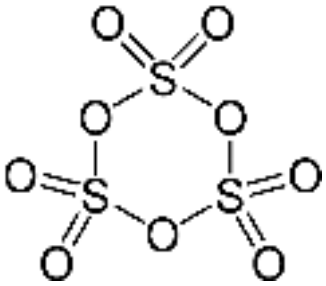
81) $Na_2B_4O_7 + H_2O \rightarrow NaOH + H_3BO_3$
Strong base Weak acid



82)



83) **Explanation -** question is asking about the number of s-s bond, s-o-s bonds, σ & π bonds in trimer of $So_3$.
**Concept -** Types & no. of bonds.
**Solution -** $(So_3)_3$ = structure



So, accordingly we can see there are
0 S-S bonds.
3 S-O-S bonds
12 σ bonds
6 π bonds (S = 0)

**Final Answer -** 2

84)

**Explanation -**
The question asks you to identify the statement that is not correct among the given options.

**Concept - (Acidic strength, Analogue, Bond length comparision)**

A. $(CH_3)_3SiOH$ is more acidic than $(CH_3)_3COH$ due to higher Si-O bond polarity and stabilization of the conjugate base by silicon.

B. Silicon monoxide (SiO), unlike stable CO, is unstable in monomeric form under normal conditions and tends to polymerize.

C. The B-F bond in $BF_3$ is shorter than in $BF_4^-$ due to pπ-pπ back bonding in $BF_3$, giving it partial double bond character.

Therefore, the statement "None of these" is incorrect because the first three statements are actually correct.

**Final Answer -** Option (4)

85)

**Explanation –** The question provides three statements about chemical bonding and molecular structures. We need to analyze each statement and determine which ones are true and which are false. The correct answer will depend on the accuracy of the statements provided.

**Concept –** The correct answer is 3. $S_1$ and $S_3$ are correct.

Here's a breakdown of why:

A. S1: Fluorine does not form any polyhalide because it does not have d-orbitals in its valence shell. This is TRUE. Fluorine's small size and lack of d-orbitals limit its ability to expand its octet and form compounds with more than one halogen atom. Polyhalides require the central atom to accommodate more than eight electrons, which fluorine cannot do.

B. S2: In $ClF_3$, the three lone pairs of electrons occupy the equatorial position. This is FALSE. $ClF_3$ has a T-shaped structure. The central chlorine atom has three bonded fluorine atoms and two lone pairs. To minimize repulsion, the lone pairs occupy equatorial positions in the T-shape, not all three.

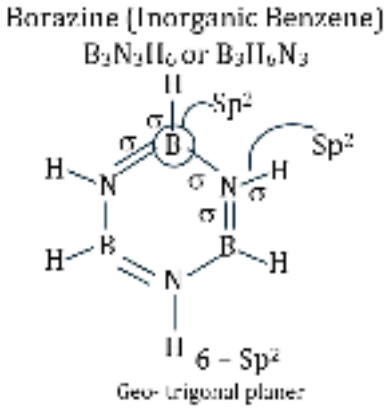
C. S3: $B_2Cl_6$ does not exist. This is TRUE. Boron, like fluorine, does not have d-orbitals readily available for bonding. It prefers to form three bonds, and $BCl_3$ is the stable form. While diborane ($B_2H_6$) exists, it has a different structure involving bridging hydrogens, not direct B-B or Cl-Cl bonds. $B_2Cl_6$ is not stable due to steric hindrance and the inability of boron to form the required number of bonds.

86)

**Question Explanation –** The question asks about the types and numbers of bonds present in inorganic benzene (borazine, $B_3N_3H_6$).

**Concept –**

A. 6 ($sp^2$-$sp^2$) σ bonds form the ring structure.
B. 6 ($sp^2$-s) σ bonds connect the B and N atoms to the hydrogen atoms.
C. 3 pπ–pπ coordinate bonds. The nitrogen donates a lone pair to the empty p orbital of the boron.




**Final Answer –** Option(4)

87) $XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$
$XeF_2 + H_2O \rightarrow Xe + 2HF + O_2$

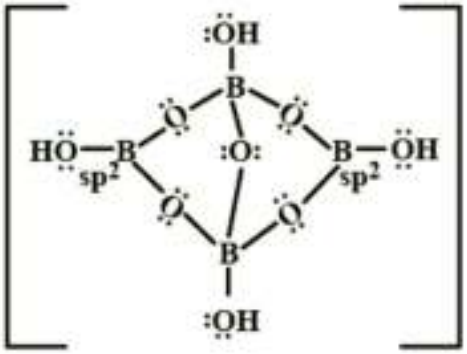
88)

**Explanation:**
We need to find the ratio of $sp^2$ and $sp^3$ hybridized boron atoms in Borax.

**Concept:**
The correct answer is 2 : 2 (or simplified to 1:1, though that's not an option).

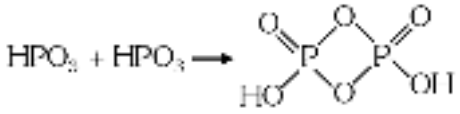
Here's why:

In the borax structure ($Na_2B_4O_7 \cdot 10H_2O$), there are four boron atoms. Two of these boron atoms are $sp^2$ hybridized and two are $sp^3$ hybridized. The borax anion contains two $BO_4$ tetrahedra ($sp^3$ boron) and two $BO_3$ triangles ($sp^2$ boron).
Therefore, the ratio of $sp^2$ to $sp^3$ hybridized boron atoms is 2 : 2.

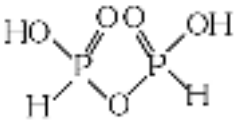


**Answer : Option (2)**

89) (i) Dimetaphosphoric acid

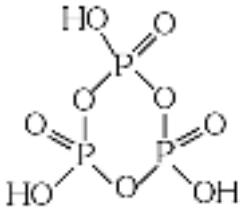


(ii) 2 mole $H_3PO_3 \rightarrow H_4P_2O_5$



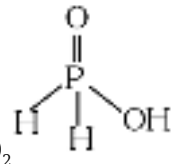
pyrophosphorous acid

(iii)



Cyclic trimetaphosphoric acid

(iv) Hypophosphorous acid - $H_3PO_2$



90)

**Generated by Allie**

**Problem Statement:** The question asks to identify which statements about graphite are correct from a given list. It tests understanding of graphite's structural properties, bonding, uses, and physical characteristics compared to diamond.

**Underlying Concept:** The key concepts involve the atomic structure of graphite, its bonding, physical properties like bond lengths and density, and its practical applications such as lubrication. Graphite consists of layers of carbon atoms bonded in an spÂ² hybridization, forming a hexagonal lattice. These layers are held together by weak van der Waals forces, making it soft and useful as a lubricant. Comparisons with diamond highlight differences in bond type, density, and structure.

**Tips and Tricks:** Remember that graphite has spÂ² hybridization creating planar layers, while diamond has spÂ³ hybridization forming a 3D network. The interlayer distance in graphite is much larger than the C-C bond length within layers. Also, graphite is generally less dense than diamond.

**Common Mistakes:** Students often confuse the interlayer spacing with bond lengths or assume graphite cannot be used at high temperatures because it is a lubricant. Another typical mistake is thinking graphite is denser than diamond.

**Why Other Options Are Incorrect?:** 1. 'C-C bond distance is nearly equal to distance between adjacent layers' is incorrect because the C-C bond length (~1.42 Ã…) is much smaller than the interlayer spacing (~3.35 Ã…). 2. 'It can be used as a lubricant but not at very high

temperature' is false because graphite remains an effective lubricant even at high temperatures in certain environments. 3. 'It has more density than diamond' is wrong because graphite is less dense (~2.2 g/cmÂ³) compared to diamond (~3.5 g/cmÂ³).

BIOLOGY

91)

i and v

92) NCERT Pg. # 130

93)

Both 1 & 3

94)

Starts in human RBCs and complete in mosquito stomach

95)

Fever includes in $1^{st}$ line of defence

96)

Ascariasis

97)

A and D

98)

A and C

99)

two

100) NCERT Pg. # 150,151

101) NCERT Page # 146 (E)

102) Module-1, Pg. # 163, 164, 165, 166

103) Module-1 Pg # 175, 176

104)

II, IV, VII and IX

105)

(i) Salmonella typhi bacteria are responsible for the infection.
(iii) By seeing the report which shows the positive result of Widal test.

106)

The exaggerated response of the immune system to certain allergen present in the environment is called allergy.

107)

Common round worm

108)

a, b & c only

109)

Sol : 2 → 1 → 4 → 3 → 5 → 6 → 8 → 7

110)

NCERT XII Pg # 134 (E), 147 (H)

111)

Sol : 4

112) NCERT-12th Pg No. # 151

113) NCERT (XII) Pg. # 146, para-2

114)

Sol : 3

115)

Sol : A, B, C, D, E

116)

drinking water containing eggs of Ascaris

117)

Microsporum - Ring worm

118) NCERT-XII, Pg. # 154

119)

Cellular barrier of innate immunity

120)

preformed antibodies

121)

Antigen

122)

Sol : 3

123)

NCERT-XII, Pg. # 136

124) NCERT Pg. # 63

125) **Explanation:**
The molecule described ($H_2L_2$) is an **antibody,** also known as immunoglobulin. Antibodies are produced by B lymphocytes and are a key component of the **humoral immune response.**

**Correct Answer: 4.** Cell-mediated immunity (CMI)

126)

One

127) NCERT-XII Pg. # 151

128) NCERT XII Pg. # 151 Para-8.2.2

129)

Difficulty in respiration, fever, chills, cough, headache

130)

All STD are completely curable

131)

NCERT XII Pg. 63

132)

Four

133)

Interferons prevent the neighbouring cells from further viral infection

134)

(A) – Disulphide bonds,
(B) – Antigen binding site,
(C) – Light chain, (D) – Heavy chain

135)

spleen and GALT

136) a → 2, b → 1, c → 1, d → 3

137) If Assertion is true statement but Reason is false

138) Both the alleles are expressed equally

139) a - 2, b - 4, c - 5, d - 3, e - 1

140) Phenotypes - 4; genotypes - 9

141) Form one linkage group

142) None of the above

143) 62, 2

144)

Both genes & chromosomes

145) Both statement I and statement II are false

146) Male have one X-chromosome and female have two X-chromosomes

147) a → 4, b → 2, c → 1, d → 3

148) Independent assortment

149) 8, 27

150) One of the parental phenotype is expressed in $F_1$ individuals

151) 21

152) 1 : 1 : 2 :3 : 3 : 6

153) 6/16

154) 34

155) 5 map units

156) 2, 4

157) D A C B

158) Law of dominance

159) Linkage map

160) ++ 40% : ab 40% : a 10% : + b 10%

161) $\frac{1}{32}$

162) B & C are correct

163) (A) $G_1$ (B) $G_2$
(C) Anaphase-l (D) Anaphase-ll

164) If Assertion is true statement but Reason is false

165) If both Assertion & Reason are true and the reason is the correct explanation of the assertion

166) 50% 100%

167) 50%

168) (i) Birds, (ii) Grasshopper

169) A: The sex determination in honey bee is based on the number of sets of chromosomes an individual receives.
B: In human beings, Y chromosome carries Sry gene.

170) Multiple alleles

171) All the seeds produced by this plant will always be wrinkled

172) 128

173) Frequency of ttRr genotype = 25%

174) 12.5%

175) Y- chromosomes

176) Mitochondria

177) Law of Segregation does not apply in this experiment

178) Starch synthesis in pea : Multiple alleles

179) 15, 7

180) At the same locus of the chromosomes